

Bai–Perron Theory Workbook: Conditional-Design Rebuild

Profile Sieve Empirical Likelihood with Nuisance Breaks

One-break estimated-partition Wilks, unknown-order selection, and proof map

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1 Primitive Setup and Notation

For a scalar array $\mathbf{a} = (a_1, \dots, a_T)'$, write

$$\|\mathbf{a}\|_T^2 = T^{-1} \sum_{t=1}^T a_t^2, \quad \|\mathbf{a}\|_\infty = \max_{t \leq T} |a_t|.$$

The empirical norm is the Euclidean geometry of the sample written as $L^2(P_T)$. Throughout the proof, $P_T = T^{-1} \sum_{t=1}^T \delta_t$ and

$$\langle \mathbf{a}, \mathbf{b} \rangle_T = T^{-1} \mathbf{a}' \mathbf{b}.$$

Thus $\|\cdot\|_T$ is the norm induced by the empirical inner product. This notation is not cosmetic: the profile step below is an exact projection KKT statement in $L^2(P_T)$. All $O_p(\cdot)$ and $o_p(\cdot)$ statements for vectors and matrices use Euclidean and spectral norm unless another norm is displayed.

The maintained null model is

$$Y_t = \beta_0 X_{t-1} + \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\} + U_t, \quad (1.1)$$

where

$$0 = k_{0,0} < k_{1,0} < \dots < k_{q_0,0} < k_{q_0+1,0} = T.$$

Throughout the known-partition theory, q_0 is fixed. The main estimated-partition theorem later specializes to one unknown nuisance break, $q_0 = 1$. Candidate searches use a trimming constant ϵ , while the true partition has slack $\epsilon_0 > \epsilon$:

$$\min_{0 \leq j \leq q_0} (k_{j+1,0} - k_{j,0}) \geq \epsilon_0 T, \quad (\bar{q} + 1)\epsilon < 1.$$

The coefficient β_0 is stable. All structural breaks are in the nuisance component.

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, let $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}, \quad j_\Lambda(t) = j \text{ if } t \in I_j(\Lambda).$$

For the search trimming constant, define the admissible candidate set

$$\mathcal{L}_q(\epsilon) = \left\{ \Lambda = (k_1, \dots, k_q) : 0 = k_0 < k_1 < \dots < k_q < k_{q+1} = T, \min_{0 \leq j \leq q} (k_{j+1} - k_j) \geq \epsilon T \right\}.$$

Let

$$\mu_t = m_{j_{\Lambda_0}(t)}(W_{t-1}), \quad \boldsymbol{\mu} = (\mu_1, \dots, \mu_T)'.$$

Let $\mathbf{P}$ be the $T \times K$ sieve matrix with t -th row

$$\mathbf{P}_K(W_{t-1})' = (p_1(W_{t-1}), \dots, p_K(W_{t-1})).$$

For a partition Λ , let $\mathbf{D}_j(\Lambda)$ be the $T \times T$ diagonal selector matrix with t -th diagonal entry $\mathbf{1}\{t \in I_j(\Lambda)\}$, $j = 0, \dots, q$. Define

$$\mathbf{Q}_{K,\Lambda} = [\mathbf{D}_0(\Lambda)\mathbf{P}, \dots, \mathbf{D}_q(\Lambda)\mathbf{P}], \quad \mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{Q}_{K,\Lambda}(\mathbf{Q}_{K,\Lambda}'\mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}_{K,\Lambda}'. \quad (1.2)$$

The Moore–Penrose inverse makes $\mathbf{M}_{K,\Lambda}$ algebraically defined even when $\mathbf{Q}'_{K,\Lambda}\mathbf{Q}_{K,\Lambda}$ is singular. Such singularity can arise if a candidate segment is too short relative to K or if the selected basis columns are collinear on that segment; the projection remains defined, but the asymptotic regularity used below may fail. The theory therefore relies on minimum-spaced admissible partitions and Gram-stability/full-rank conditions on the relevant candidate sets.

For each (β, Λ) , define

$$\hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{M}_{K,\Lambda}(\mathbf{Y} - \beta \mathbf{X}_{lag}), \quad (1.3)$$

$$\mathbf{w}^c(\Lambda) = \mathbf{M}_{K,\Lambda} \mathbf{w}, \quad (1.4)$$

$$\hat{Z}_t(\beta, \Lambda) = \hat{u}_t(\beta, \Lambda) w_t^c(\Lambda). \quad (1.5)$$

Lemma 1 (Sample $L^2(P_T)$ profile KKT). *Fix (β, Λ) , write $\mathbf{Q} = \mathbf{Q}_{K,\Lambda}$, and equip $\mathbb{R}^T$ with the empirical inner product $\langle \mathbf{a}, \mathbf{b} \rangle_T = T^{-1} \mathbf{a}' \mathbf{b}$. The fitted value $\mathbf{Q} \hat{\mathbf{c}}$ from*

$$\min_{\mathbf{c}} \|\mathbf{Y} - \beta \mathbf{X}_{lag} - \mathbf{Q} \mathbf{c}\|_T^2$$

is the $L^2(P_T)$ orthogonal projection of $\mathbf{Y} - \beta \mathbf{X}_{lag}$ onto $\text{col}(\mathbf{Q})$. Its residual

$$\hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{M}_{K,\Lambda}(\mathbf{Y} - \beta \mathbf{X}_{lag})$$

satisfies the KKT normal equations

$$\mathbf{Q}' \hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{0}.$$

Likewise, $\mathbf{w}^c(\Lambda) = \mathbf{M}_{K,\Lambda} \mathbf{w}$ is the residual from the $L^2(P_T)$ projection of $\mathbf{w}$ onto $\text{col}(\mathbf{Q})$, and

$$\mathbf{Q}' \mathbf{w}^c(\Lambda) = \mathbf{0}.$$

Proof. The objective is a convex quadratic. Its first-order normal equation is $\mathbf{Q}'(\mathbf{Y} - \beta \mathbf{X}_{lag} - \mathbf{Q} \mathbf{c}) = \mathbf{0}$. When $\mathbf{Q}' \mathbf{Q}$ is singular, the coefficient vector need not be unique, but the fitted value is the Moore–Penrose projection

$$\mathbf{Q}(\mathbf{Q}' \mathbf{Q})^\dagger \mathbf{Q}'(\mathbf{Y} - \beta \mathbf{X}_{lag}),$$

and the residual is $\mathbf{M}_{K,\Lambda}(\mathbf{Y} - \beta \mathbf{X}_{lag})$. Hence $\mathbf{Q}' \hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{0}$. The same argument with $\mathbf{w}$ in place of $\mathbf{Y} - \beta \mathbf{X}_{lag}$ gives $\mathbf{Q}' \mathbf{w}^c(\Lambda) = \mathbf{0}$. These two normal equations are the finite-sample source of the nuisance annihilation used later: any component in $\text{col}(\mathbf{Q})$ is orthogonal to both the profiled residual and the residualized score weight. $\square$

Let

$$S_T(\Lambda) = T^{-1/2} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda), \quad V_T(\Lambda) = T^{-1} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda)^2.$$

For a scalar score array $Z_1, \dots, Z_T$ satisfying the convex-hull condition, let λ_T be the unique solution of

$$\sum_{t=1}^T \frac{Z_t}{1 + \lambda_T Z_t} = 0, \quad 1 + \lambda_T Z_t > 0 \quad (t \leq T),$$

and define the twice negative empirical-likelihood log ratio by

$$\ell_T(Z_1, \dots, Z_T) = 2 \sum_{t=1}^T \log(1 + \lambda_T Z_t). \quad (1.6)$$

In particular,

$$\ell_T(\beta, \Lambda) = \ell_T\{\widehat{Z}_1(\beta, \Lambda), \dots, \widehat{Z}_T(\beta, \Lambda)\}.$$

Remark 1 (Three KKT systems). The proof uses three linked KKT systems. First, nuisance profiling solves a quadratic projection KKT problem in $L^2(P_T)$, producing $\widehat{\mathbf{u}}(\beta, \Lambda)$ and $\mathbf{w}^c(\Lambda)$ in the empirical orthogonal complement of the sieve space. Second, empirical likelihood solves an entropy KKT problem on the sample simplex, producing the multiplier λ_T . Third, as empirical inner products converge to their population limits, the sample normal-score equation approaches a population $L^2(P)$ KKT equation. Identification comes from uniqueness of the population zero; Wilks comes from the local $L^2(P_T)$ curvature of the EL KKT plus the conditional CLT and LLN for the oracle score.

Remark 2 (Deterministic geometry on good events). Apart from the oracle CLT, denominator LLN, maximum-score bound, and the probability of the required good events, the proof is deterministic geometry. On events where the Gram matrices are stable, the sieve envelopes hold, the candidate partition is localized, and the EL convex-hull qualification holds, the required bounds follow from projection contraction, Pythagoras, Cauchy–Schwarz, support size of the boundary block, and finite-dimensional KKT calculus. The stochastic lemmas below therefore verify the inputs to this algebra; they are not the organizing principle of the proof. In the one-break bridge this bookkeeping is collected explicitly in Lemma ??.

For an interval $[s, e]$, define the null-imposed segment cost

$$\mathcal{C}_K(s, e; \beta) = \min_{\mathbf{c} \in \mathbb{R}^K} \sum_{t=s}^e \{Y_t - \beta X_{t-1} - \mathbf{P}_K(W_{t-1})' \mathbf{c}\}^2.$$

For $\Lambda = (k_1, \dots, k_q)$, define

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^q \mathcal{C}_K(k_j + 1, k_{j+1}; \beta).$$

All minimizers are measurable selections with a fixed deterministic tie-breaking rule. For known order,

$$\widehat{\Lambda}_q(\beta) = \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda), \quad \widetilde{\Lambda} = \widehat{\Lambda}_{q_0}(\beta_0).$$

For unknown order,

$$\widehat{q}(\beta) = \arg \min_{0 \leq q \leq \bar{q}} \{RSS_K(\beta, \widehat{\Lambda}_q(\beta)) + \text{pen}_T(q, K)\}, \quad \widehat{\Lambda}(\beta) = \widehat{\Lambda}_{\widehat{q}(\beta)}(\beta).$$

Although the Moore–Penrose inverse makes $\mathbf{M}_{K, \Lambda}$ algebraically defined for arbitrary Λ , all theoretical searches are restricted to minimum-spaced admissible partitions $\mathcal{L}_q(\epsilon)$; thus no

candidate segment is allowed to be asymptotically shorter than the sieve dimension K , and the Gram matrices used in the proofs are nonsingular on the relevant candidate set.

The sieve dimension is allowed to depend on the sample size and is written $K = K_T$. Throughout, $K_T \rightarrow \infty$, subject to the rate restrictions in Assumptions 3–6 and the local Gram regularity conditions below. The basis is normalized so that its first component is constant, say $p_1(w) \equiv 1$, allowing regime-specific intercept and level shifts to be represented by the block sieve.

For local-power calculations, distinguish the null value used in the statistic from the data-generating slope. The local alternative used in the rebuilt theorem is

$$\beta_T = \beta_0 + \frac{h}{\sqrt{T}}, \quad h \in \mathbb{R},$$

with data generated under P_{β_T} while the profile empirical likelihood is evaluated at β_0 . Equivalently,

$$Y_t - \beta_0 X_{t-1} = \mu_t + U_t + \frac{h}{\sqrt{T}} X_{t-1},$$

with the same nuisance break partition Λ_0 . Section 8 records the corresponding drift conditions.

2 Primitive Assumptions

Assumption 1 (Conditional-on-design stochastic environment). *Let*

$$\mathcal{D}_T = \sigma\{X_{t-1}, W_{t-1}, w_t : 1 \leq t \leq T\}.$$

The inclusion of w_t is deliberate: the residualized empirical-likelihood weight $\mathbf{M}_{K,\Lambda}\mathbf{w}$ must be fixed in the conditional-design arguments. Conditional on $\mathcal{D}_T$, the variables $U_{1,T}, \dots, U_{T,T}$ are independent and satisfy

$$\mathbb{E}(U_{t,T} \mid \mathcal{D}_T) = 0, \quad 0 < \underline{\sigma}^2 \leq \sigma_{t,T}^2 := \mathbb{E}(U_{t,T}^2 \mid \mathcal{D}_T) \leq \bar{\sigma}^2 < \infty.$$

Moreover, for every $s \in \mathbb{R}$,

$$\mathbb{E}\{\exp(sU_{t,T}) \mid \mathcal{D}_T\} \leq \exp(Cs^2/2)$$

almost surely, uniformly in t and T . The subscript T is suppressed below when no ambiguity can arise.

Assumption 2 (Design-measurable score regularity). *For every admissible partition Λ , the matrices $\mathbf{Q}_{K,\Lambda}$, $\mathbf{M}_{K,\Lambda}$, $\mathbf{\Pi}_{K,\Lambda} = \mathbf{I}_T - \mathbf{M}_{K,\Lambda}$, and $\mathbf{v}_\Lambda = \mathbf{M}_{K,\Lambda}\mathbf{w}$ are $\mathcal{D}_T$ -measurable. Hence the full-sample residualized instrument creates no predictability problem: all oracle score calculations are conditional-on-design calculations. For the true partition, write $\mathbf{v}_0 = \mathbf{v}_{\Lambda_0}$.*

Assumption 3 (Bounded weights, nuisance functions, and basis envelope). *The sieve dimension satisfies $K = K_T \rightarrow \infty$. The first basis element is constant, $p_1(w) \equiv 1$. There are constants $C_w, C_m < \infty$ and an envelope ζ_K such that*

$$\max_{t \leq T} |w_t| \leq C_w, \quad \max_{0 \leq j \leq q_0} \sup_x |m_j(x)| \leq C_m, \quad \max_{t \leq T} \|\mathbf{P}_K(W_{t-1})\| \leq \zeta_K$$

with probability tending to one.

Assumption 4 (Spacing, smoothness, and break detectability). *The known-partition theory allows fixed q_0 . The estimated-partition and unknown-order Wilks theorems in the main text impose $q_0 = 1$. The true segments obey the strict spacing condition in the setup with $\epsilon_0 > \epsilon$, and searches are over $\mathcal{L}_q(\epsilon)$ for $0 \leq q \leq \bar{q}$. Each m_j belongs to a common smoothness class $\mathcal{H}^s$. If $q_0 \geq 1$, define the design jump size by*

$$\Delta_T = \min_{0 \leq j < q_0} \left\{ T^{-1} \sum_{t=1}^T [m_{j+1}(W_{t-1}) - m_j(W_{t-1})]^2 \right\}^{1/2}.$$

For the one-break main theorem, this reduces to the empirical jump norm between m_0 and m_1 , and $\Delta_T > 0$.

Assumption 5 (Global and local Gram stability). *Uniformly over every interval J that can appear as a segment in an admissible search with $|J| \geq \epsilon T$,*

$$c_G I_K \preceq |J|^{-1} \mathbf{P}'_J \mathbf{P}_J \preceq C_G I_K$$

with probability tending to one. The same bounds hold on all one-break local segments satisfying $|k - k_0| \leq Cr_T$. This condition is stronger than algebraic Moore–Penrose existence: it supplies the inverse and leverage bounds used by the break bridge and by the profile-RSS search.

Assumption 6 (Sieve approximation and projection bias). *For each regime j , there is a coefficient vector $\mathbf{c}_{j,K}$ such that*

$$m_j(W_{t-1}) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_{j,K} + r_{j,K,t}, \quad \max_j \|\mathbf{r}_{j,K}\|_T = O_p(a_K), \quad \max_j \|\mathbf{r}_{j,K}\|_\infty = O_p(a_K).$$

Let $\mathbf{r}_0$ be the stacked true-partition approximation error. The rates satisfy

$$T^{-1/2} \mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w} = o_p(1), \quad a_K(1 + \zeta_K) \rightarrow 0, \quad a_K(1 + \zeta_K)^2 = o(\sqrt{T}).$$

Also,

$$\frac{(q_0 + 1)K}{\sqrt{T}} \rightarrow 0, \quad \zeta_K \sqrt{\frac{(q_0 + 1)K}{T}} \rightarrow 0.$$

Assumption 7 (Segmentwise orthogonality route for score bias). *For each true segment $I_{j,0}$, choose $\mathbf{c}_{j,K}$ as the empirical least-squares projection of $m_j(W_{t-1})$ on $\mathbf{P}_K(W_{t-1})$ within that segment, so*

$$\mathbf{P}'_{j,0} \mathbf{r}_{j,K} = 0.$$

The instrument admits a segmentwise sieve approximation

$$\mathbf{w}_{j,0} = \mathbf{P}_{j,0} \boldsymbol{\gamma}_{j,K} + \mathbf{s}_{j,K}, \quad \max_j \|\mathbf{s}_{j,K}\|_T = O_p(c_K),$$

where $\|\cdot\|_T$ uses the full-sample normalization. The rates satisfy

$$a_K \rightarrow 0, \quad \sqrt{T} a_K c_K \rightarrow 0.$$

Lemma 2 (Primitive score-bias verification). *Under Assumptions 5, 6 without its first score-bias display, and 7,*

$$T^{-1/2} \mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w} = o_p(1).$$

Thus the score-bias display in Assumption 6 is discharged by segmentwise empirical projection and a smooth residualized instrument.

Detailed solution. On the true partition, $\mathbf{M}_{K, \Lambda_0}$ is block diagonal across the true segments. Let $\mathbf{M}_{j,0}$ be the residual-maker for $\mathbf{P}_{j,0}$ on segment j . Since $\mathbf{P}'_{j,0} \mathbf{r}_{j,K} = 0$, $\mathbf{M}_{j,0} \mathbf{r}_{j,K} = \mathbf{r}_{j,K}$. Also $\mathbf{M}_{j,0} \mathbf{w}_{j,0} = \mathbf{M}_{j,0} \mathbf{s}_{j,K}$. Hence

$$|\mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w}| \leq \sum_{j=0}^{q_0} \|\mathbf{r}_{j,K}\| \|\mathbf{M}_{j,0} \mathbf{s}_{j,K}\| \leq \sum_{j=0}^{q_0} \|\mathbf{r}_{j,K}\| \|\mathbf{s}_{j,K}\|.$$

Because q_0 is fixed and the norms in Assumptions 6 and 7 use full-sample normalization, the last sum is $O_p(T a_K c_K)$. Multiplication by $T^{-1/2}$ gives $O_p(\sqrt{T} a_K c_K) = o_p(1)$. $\square$

Assumption 8 (One-break localization input for the bridge). *In the one-break bridge, $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\tilde{\Lambda} = (\tilde{k})$. The bridge uses*

$$|\tilde{k} - k_0| = O_p(r_T), \quad r_T = o(\sqrt{T}).$$

Section 7 derives this rate from the conditional-design profile-sieve segmentation conditions with $r_T = R_T / \Delta_T^2$.

Assumption 9 (Empirical-likelihood convex hull). *For the score array under consideration,*

$$\mathbb{P}\{\min_{t \leq T} Z_{t,T} < 0 < \max_{t \leq T} Z_{t,T}\} \rightarrow 1.$$

When used for the true partition, $Z_{t,T} = \hat{Z}_t(\beta_0, \Lambda_0)$. When used for the estimated partition, $Z_{t,T} = \hat{Z}_t(\beta_0, \tilde{\Lambda})$.

Assumption 10 (Primitive sign mass for convex hull). *There are constants $c_u, c_v, p_0, \pi > 0$ and design-measurable sets $J_T \subset \{1, \dots, T\}$ with $|J_T| \geq \pi T$ such that*

$$|v_{t,0}| \geq c_v, \quad t \in J_T,$$

and, conditionally on $\mathcal{D}_T$,

$$\mathbb{P}(U_t > c_u \mid \mathcal{D}_T) \geq p_0, \quad \mathbb{P}(U_t < -c_u \mid \mathcal{D}_T) \geq p_0, \quad t \in J_T.$$

For the estimated one-break score, the feasible-score perturbation away from $O_p(r_T)$ boundary observations satisfies

$$a_K b_K^2 + b_K \zeta_K \sqrt{\frac{\log\{T(1+r_T)\}}{T}} + \frac{b_K r_T \zeta_K^2}{T} \rightarrow 0. \quad (2.1)$$

Lemma 3 (Primitive convex-hull verification). *Under Assumptions 1, 3, 5, 6, 11, and 10, Assumption 9 holds for the true partition score. If, in addition, $q_0 = 1$, $|\tilde{k} - k_0| = O_p(r_T)$, and the local bridge rates hold, then Assumption 9 also holds for the estimated one-break score.*

Detailed solution. Condition on $\mathcal{D}_T$. Let $J_T^+ = \{t \in J_T : v_{t,0} \geq c_v\}$ and $J_T^- = \{t \in J_T : v_{t,0} \leq -c_v\}$. At least one of these sets has cardinality at least $\pi T/2$. If $|J_T^+| \geq \pi T/2$, then a positive oracle score occurs when some $U_t > c_u$ on J_T^+ , and a negative oracle score occurs when some $U_t < -c_u$ on J_T^+ . Conditional independence bounds the probability of missing either sign by $2(1 - p_0)^{\pi T/2}$. The case $|J_T^-| \geq \pi T/2$ is identical with the signs reversed. For the true partition, the sharper leverage form of the projection-noise bound and the sieve approximation bound imply a uniform $o_p(1)$ feasible-score perturbation on J_T ; this is the $r_T = 0$ case of the transfer rate below. Thus both signs remain present for $\hat{Z}_t(\beta_0, \Lambda_0)$.

For the estimated partition, remove the boundary set where assignments differ; Lemma 13 leaves at least $\pi T - o_p(T)$ observations. On the remaining observations, the coefficient-update bound, the uniform projection-noise bound, and the sieve approximation bound give the transfer rate in (2.1). Hence the estimated score has both signs with probability tending to one. $\square$

3 Conditional-Design Completing Conditions

The rebuilt proof uses one stochastic environment throughout. Conditional on $\mathcal{D}_T$, the sieve design, all projection matrices, residualized weights, and break estimators are fixed. The remaining conditions below are rate and identification conditions; they do not reintroduce a predictable-score route.

Write

$$d_K = (q_0 + 1)K, \quad b_K = 1 + \zeta_K.$$

Assumption 11 (Oracle envelope completion). *For the true partition,*

$$\max_{t \leq T} |v_{t,0}| = O_p(b_K), \quad \frac{b_K^2 \log T}{T} \rightarrow 0.$$

The first display follows from the leverage bound under Assumptions 3 and 5; it is stated here to make the known-partition Wilks theorem self-contained. The logarithmic rate is the conditional sub-Gaussian maximum-score rate used below.

Assumption 12 (Local sieve-bias stability). *In the one-break section, for every fixed $C < \infty$,*

$$\sup_{|k - k_0| \leq Cr_T} T^{-1/2} |\mathbf{r}'_0 \{\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0}\} \mathbf{w}| = o_p(1). \quad (3.1)$$

A sufficient primitive rate, proved by the coefficient-update argument in Lemma 15, is

$$\frac{a_K b_K (1 + \zeta_K^2) r_T}{\sqrt{T}} \rightarrow 0. \quad (3.2)$$

Assumption 13 (Local full-rank series regularity). *In the one-break section, every segment Gram matrix associated with a partition satisfying $|k - k_0| \leq Cr_T$ is nonsingular, and its eigenvalues are between cT and CT , uniformly with probability tending to one. Also*

$$\zeta_K^2 = O(K). \quad (3.3)$$

The last display is the usual normalized-series envelope condition. It is used to simplify the denominator bridge rate; without it the larger order $b_K^2 \zeta_K^2 r_T / T$ should be retained.

Assumption 14 (Deterministic conditional-design segmentation primitives). *This is the deterministic part of the route used to discharge the profile-sieve break search in the one-break main theorem. Let $\boldsymbol{\mu}_K = \mathbf{Q}_{K, \Lambda_0} \mathbf{c}_0$ be the sieve part of the true nuisance mean and define the deterministic conditional profile loss*

$$D_T(\Lambda) = \|\mathbf{M}_{K, \Lambda} \boldsymbol{\mu}_K\|^2.$$

At the true partition, $D_T(\Lambda_0) = 0$. For one-break candidates,

$$D_T(k) \geq c_\Delta |k - k_0| \Delta_T^2.$$

For underfitted candidates $q < q_0$,

$$\inf_{\Lambda \in \mathcal{L}_q(\epsilon)} D_T(\Lambda) \geq c_\Delta T \Delta_T^2.$$

The approximation error obeys the sup-norm rate in Assumption 6; write

$$R_T = K + \log T + T a_K^2, \quad r_T = R_T / \Delta_T^2.$$

The rates satisfy

$$R_T = o(T \Delta_T^2), \quad r_T = o(\sqrt{T}).$$

Together with Assumptions 1, 5, and 6, Proposition 2 verifies the profile-RSS localization and the overfit/underfit RSS gaps recorded in Assumption 16.

Remark 3 (Why the rebuild uses conditional design). The former martingale version needed a predictable approximation to the full-sample residualized instrument. The conditional-design route avoids that problem: $\mathbf{M}_{K, \Lambda}$, $\mathbf{v}_\Lambda$, and the RSS minimizer are all $\mathcal{D}_T$ -measurable, so stochastic arguments are conditional concentration and conditional triangular-array CLT arguments. The cost is a stronger exogeneity assumption and, in the main estimated-break Wilks theorem, the explicit one-break scope $q_0 = 1$.

For a scalar score array $Z_1, \dots, Z_T$, empirical likelihood is not a projection problem. It is the entropy KKT problem on the sample simplex

$$\max_{\pi_1, \dots, \pi_T} \sum_{t=1}^T \log(T\pi_t) \quad \text{subject to} \quad \pi_t > 0, \quad \sum_{t=1}^T \pi_t = 1, \quad \sum_{t=1}^T \pi_t Z_t = 0.$$

Its connection with $L^2(P_T)$ appears only locally: near a small multiplier, the derivative of the EL KKT equation has curvature $-T^{-1} \sum_t Z_t^2$, the empirical $L^2(P_T)$ norm squared of the score.

Lemma 4 (Sample empirical-likelihood KKT). *If zero lies in the interior of the convex hull of $Z_1, \dots, Z_T$, the empirical-likelihood simplex program has weights*

$$\hat{\pi}_t = \frac{1}{T(1 + \lambda_T Z_t)}, \quad t = 1, \dots, T,$$

where λ_T is the unique value satisfying

$$\sum_{t=1}^T \frac{Z_t}{1 + \lambda_T Z_t} = 0, \quad 1 + \lambda_T Z_t > 0 \quad (t \leq T).$$

The twice negative empirical-likelihood log ratio is therefore

$$2 \sum_{t=1}^T \log(1 + \lambda_T Z_t).$$

Proof. The Lagrangian first-order condition gives

$$\pi_t^{-1} = \nu + \eta Z_t.$$

Multiplying by π_t and summing over t , while using $\sum_t \pi_t = 1$ and $\sum_t \pi_t Z_t = 0$, gives $\nu = T$. With $\lambda_T = \eta/T$,

$$\pi_t = \{T(1 + \lambda_T Z_t)\}^{-1}.$$

Substitution into the moment constraint gives the displayed multiplier equation. The interior convex-hull condition gives existence, positivity, and uniqueness of the scalar multiplier. $\square$

Lemma 5 (Local $L^2(P_T)$ curvature of the EL KKT). *Let $Z_{1,T}, \dots, Z_{T,T}$ satisfy the convex-hull condition and suppose*

$$S_1 = \sum_{t=1}^T Z_{t,T} = O_p(\sqrt{T}), \quad S_2 = \sum_{t=1}^T Z_{t,T}^2 \geq (c + o_p(1))T,$$

for some $c > 0$. If $M_T = \max_{t \leq T} |Z_{t,T}| = o_p(\sqrt{T})$, then the EL multiplier satisfies

$$\lambda_T = \frac{S_1}{S_2} + o_p(T^{-1/2}), \quad |\lambda_T| M_T = o_p(1).$$

Equivalently,

$$\lambda_T = \frac{T^{-1} \sum_t Z_{t,T}}{T^{-1} \sum_t Z_{t,T}^2} + o_p(T^{-1/2}).$$

The denominator in this display is the empirical $L^2(P_T)$ curvature of the EL KKT equation.

Proof. The EL KKT equation is

$$0 = \sum_{t=1}^T \frac{Z_{t,T}}{1 + \lambda_T Z_{t,T}}.$$

The convex-hull condition gives a unique multiplier on the interval where all $1 + \lambda_T Z_{t,T}$ are positive. Using $z/(1 + \lambda z) = z - \lambda z^2/(1 + \lambda z)$, the KKT equation implies

$$S_1 = \lambda_T \sum_{t=1}^T \frac{Z_{t,T}^2}{1 + \lambda_T Z_{t,T}}.$$

Consequently,

$$|S_1| \geq \frac{|\lambda_T| S_2}{1 + |\lambda_T| M_T}.$$

After rearrangement,

$$|\lambda_T| \{S_2 - |S_1| M_T\} \leq |S_1|.$$

Now $|S_1| = O_p(\sqrt{T})$, $S_2 \geq (c + o_p(1))T$, and $|S_1| M_T = o_p(T)$. Hence

$$\lambda_T = O_p(T^{-1/2}), \quad |\lambda_T| M_T = o_p(1).$$

Using $(1 + x)^{-1} = 1 - x + x^2/(1 + x)$ in the KKT equation gives

$$0 = S_1 - \lambda_T S_2 + R_T, \quad |R_T| \leq \frac{\lambda_T^2 M_T S_2}{1 - |\lambda_T| M_T} = o_p(\sqrt{T}).$$

Since S_2 is bounded below by order T , the displayed expansion for λ_T follows. $\square$

Lemma 6 (Scalar empirical-likelihood expansion). *Let $Z_{1,T}, \dots, Z_{T,T}$ satisfy the convex-hull condition and suppose*

$$A_T = T^{-1/2} \sum_{t=1}^T Z_{t,T} = O_p(1), \quad B_T = T^{-1} \sum_{t=1}^T Z_{t,T}^2 \geq c + o_p(1)$$

for some $c > 0$, and

$$\max_{t \leq T} |Z_{t,T}| = o_p(\sqrt{T}).$$

Then

$$\ell_T(Z_{1,T}, \dots, Z_{T,T}) = \frac{A_T^2}{B_T} + o_p(1). \quad (3.4)$$

Proof. Put $S_1 = \sum_t Z_{t,T}$, $S_2 = \sum_t Z_{t,T}^2$, and $M_T = \max_t |Z_{t,T}|$. Lemma 5 gives

$$\lambda_T = \frac{S_1}{S_2} + o_p(T^{-1/2}), \quad |\lambda_T| M_T = o_p(1).$$

Finally, Taylor expansion of the logarithm, uniformly because $|\lambda_T| M_T = o_p(1)$, yields

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = 2\lambda_T S_1 - \lambda_T^2 S_2 + R_{2T},$$

where

$$|R_{2T}| \leq C |\lambda_T|^3 M_T S_2 = o_p(1).$$

Substituting the expansion for λ_T gives

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = \frac{S_1^2}{S_2} + o_p(1) = \frac{A_T^2}{B_T} + o_p(1),$$

which proves (3.4). $\square$

4 Deterministic Projection Algebra

Lemma 7 (Block projection identities). *For every partition Λ ,*

$$\mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = \mathbf{0}, \quad \mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}, \quad \mathbf{M}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

Consequently, for every conformable $\mathbf{c}$,

$$(\mathbf{Q}_{K,\Lambda} \mathbf{c})' \mathbf{M}_{K,\Lambda} \mathbf{w} = 0.$$

Detailed solution. Abbreviate $\mathbf{Q} = \mathbf{Q}_{K,\Lambda}$. Take a thin singular-value decomposition

$$\mathbf{Q} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r',$$

where $r = \text{rank}(\mathbf{Q})$, the columns of $\mathbf{U}_r$ and $\mathbf{V}_r$ are orthonormal, and $\mathbf{\Sigma}_r$ is diagonal with strictly positive entries. Then

$$\mathbf{Q}'\mathbf{Q} = \mathbf{V}_r \mathbf{\Sigma}_r^2 \mathbf{V}_r', \quad (\mathbf{Q}'\mathbf{Q})^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-2} \mathbf{V}_r'.$$

Therefore

$$\mathbf{\Pi}_{K,\Lambda} = \mathbf{Q}(\mathbf{Q}'\mathbf{Q})^\dagger \mathbf{Q}' = \mathbf{U}_r \mathbf{U}_r'.$$

This is the orthogonal projector onto $\text{col}(\mathbf{Q})$. In particular,

$$\mathbf{\Pi}'_{K,\Lambda} = \mathbf{\Pi}_{K,\Lambda}, \quad \mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{\Pi}_{K,\Lambda}, \quad \mathbf{\Pi}_{K,\Lambda} \mathbf{Q} = \mathbf{Q}.$$

Since $\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}$,

$$\mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}$$

and

$$\mathbf{M}_{K,\Lambda}^2 = (\mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda})^2 = \mathbf{I}_T - 2\mathbf{\Pi}_{K,\Lambda} + \mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

Also

$$\mathbf{Q}' \mathbf{M}_{K,\Lambda} = \{\mathbf{M}_{K,\Lambda} \mathbf{Q}\}' = \mathbf{0}'.$$

For conformable $\mathbf{c}$ and $\mathbf{w}$, it follows immediately that

$$(\mathbf{Q}\mathbf{c})' \mathbf{M}_{K,\Lambda} \mathbf{w} = \mathbf{c}' \mathbf{Q}' \mathbf{M}_{K,\Lambda} \mathbf{w} = 0.$$

□

Assumption 15 (Population KKT bridge and identification). *Let $\mathcal{B}$ be a compact neighborhood of β_0 . Let $\mathcal{N}_{\Lambda_0}^P$ be the $L^2(P)$ closure of the true regime-specific nuisance space generated by $\mathbf{1}\{j_{\Lambda_0}(t) = j\}h(W_{t-1})$. Uniformly over $\beta \in \mathcal{B}$, the empirical projection moments and profiled score moments converge to their $L^2(P)$ limits, the empirical Gram matrices are stable, and the sieve approximation errors vanish at the rates in Assumption 6. Let*

$$u_\beta = Y - \beta X_{lag} - \Pi_P(Y - \beta X_{lag} \mid \mathcal{N}_{\Lambda_0}^P), \quad v = w - \Pi_P(w \mid \mathcal{N}_{\Lambda_0}^P),$$

where $\Pi_P(\cdot \mid \mathcal{N}_{\Lambda_0}^P)$ denotes $L^2(P)$ orthogonal projection. The population normal-score equation

$$\Psi(\beta) = \mathbb{E}\{u_\beta v\} = 0$$

has the unique solution $\beta = \beta_0$ in $\mathcal{B}$.

Proposition 1 (From sample KKT to population KKT). *Under Assumption 15, any sequence $\hat{\beta}_T \in \mathcal{B}$ whose sample profiled score satisfies*

$$T^{-1} \sum_{t=1}^T \hat{u}_t(\hat{\beta}_T, \hat{\Lambda}) w_t^c(\hat{\Lambda}) = o_p(1)$$

also satisfies $\Psi(\hat{\beta}_T) = o_p(1)$. Consequently,

$$\hat{\beta}_T \xrightarrow{p} \beta_0.$$

Proof. Lemma 1 puts both the profiled residual and the residualized score weight in the empirical orthogonal complement of the sample sieve space. Assumption 15 transfers these empirical normal equations to their $L^2(P)$ limits, so the sample profiled score is uniformly close to $\Psi(\beta)$ on $\mathcal{B}$. If the sample score at $\hat{\beta}_T$ is $o_p(1)$, then $\Psi(\hat{\beta}_T) = o_p(1)$. Compactness of $\mathcal{B}$, continuity of Ψ , and uniqueness of the population zero imply $\hat{\beta}_T \rightarrow_p \beta_0$. $\square$

Remark 4 (What the KKT bridge does and does not prove). The KKT bridge proves identification and consistency once uniform convergence and population uniqueness are available. It does not by itself prove Wilks. For Wilks, the EL KKT must be localized near $\lambda_T = 0$, which requires the oracle numerator CLT, denominator LLN, and maximum-score control proved below. Those stochastic lemmas are therefore not replacements for the KKT geometry; they are the conditions that make the EL KKT use its local $L^2(P_T)$ curvature.

Lemma 8 (Uniform fitted-value and leverage bounds). *Under Assumptions 3 and 5, uniformly over*

$$\mathcal{N}_T(r_T) = \{\Lambda : q = q_0, \max_{1 \leq j \leq q_0} |k_j - k_{j,0}| \leq Cr_T\},$$

$$\|\mathbf{M}_{K,\Lambda} \mathbf{w}\|_\infty = O_p(1 + \zeta_K), \quad \max_{t \leq T} \{\mathbf{Q}_{K,\Lambda} (\mathbf{Q}'_{K,\Lambda} \mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}'_{K,\Lambda}\}_{tt} = O_p(\zeta_K^2 K/T).$$

Detailed solution. Write $\mathbf{Q} = \mathbf{Q}_{K,\Lambda}$, $\mathbf{G} = \mathbf{Q}'\mathbf{Q}$, and let $\mathbf{q}'_t$ denote row t of $\mathbf{Q}$. Because exactly one regime block is active in each row,

$$\|\mathbf{q}_t\| = \|\mathbf{P}_K(W_{t-1})\| \leq \zeta_K.$$

On the uniform Gram event in Assumption 5,

$$\|\mathbf{G}^\dagger\| \leq \frac{C}{T}, \quad \|\mathbf{Q}\|^2 = \lambda_{\max}(\mathbf{Q}'\mathbf{Q}) \leq CT.$$

Since $\|\mathbf{w}\| \leq C_w \sqrt{T}$, the least-squares coefficient vector

$$\hat{\mathbf{c}}_\Lambda = \mathbf{G}^\dagger \mathbf{Q}' \mathbf{w}$$

satisfies

$$\|\hat{\mathbf{c}}_\Lambda\| \leq \frac{C}{T} \|\mathbf{Q}\| \|\mathbf{w}\| \leq C.$$

Thus, uniformly in t and Λ ,

$$|(\mathbf{\Pi}_{K,\Lambda} \mathbf{w})_t| = |\mathbf{q}'_t \hat{\mathbf{c}}_\Lambda| \leq C \zeta_K,$$

and hence

$$\|\mathbf{M}_{K,\Lambda}\mathbf{w}\|_\infty \leq \|\mathbf{w}\|_\infty + \|\mathbf{\Pi}_{K,\Lambda}\mathbf{w}\|_\infty = O_p(1 + \zeta_K).$$

For the leverage,

$$(\mathbf{\Pi}_{K,\Lambda})_{tt} = \mathbf{q}_t' \mathbf{G}^\dagger \mathbf{q}_t \leq \|\mathbf{G}^\dagger\| \|\mathbf{q}_t\|^2 \leq C \frac{\zeta_K^2}{T}.$$

This is sharper than the displayed target and therefore implies $O_p(\zeta_K^2 K/T)$, because $K \geq 1$. All inequalities hold on a single uniform Gram event whose probability tends to one. $\square$

5 Known-Partition Conditional-Design Wilks

Set

$$\mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0}\mathbf{w}, \quad A_T^{(q)} = T^{-1/2} \sum_{t=1}^T U_t v_{t,0}, \quad V_T^{(q)} = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2.$$

Lemma 9 (Oracle conditional-design CLT). *Under Assumptions 1–5 and 11, suppose*

$$\eta_{T,q}^2 := T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad \mathbb{P}(\eta_q \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$. Then

$$A_T^{(q)} \xrightarrow{\text{st}} \eta_q Z,$$

where $Z \sim N(0, 1)$ is independent of the limiting sigma-field generated by η_q .

Detailed solution. Condition on $\mathcal{D}_T$. The summands $T^{-1/2}U_t v_{t,0}$ are independent, centered, and have conditional variance $T^{-1}\sigma_t^2 v_{t,0}^2$. The conditional variance sum is $\eta_{T,q}^2$. The conditional sub-Gaussian envelope and $\max_t |v_{t,0}| = O_p(b_K)$, together with $b_K/\sqrt{T} \rightarrow 0$, imply the Lindeberg condition conditionally on $\mathcal{D}_T$: for every $\varepsilon > 0$,

$$\sum_{t=1}^T \mathbb{E} \left[T^{-1} U_t^2 v_{t,0}^2 \mathbf{1}\{|U_t v_{t,0}| > \varepsilon \sqrt{T}\} \mid \mathcal{D}_T \right] = o_p(1).$$

Therefore the conditional triangular-array CLT gives

$$\mathbb{E}\{\exp(iu A_T^{(q)}) \mid \mathcal{D}_T\} - \exp(-u^2 \eta_{T,q}^2/2) \xrightarrow{p} 0$$

for every fixed u . Since $\eta_{T,q}^2 \rightarrow_p \eta_q^2$, the usual conditional-characteristic-function argument yields stable convergence to $\eta_q Z$. The lower bound on η_q is recorded for the later studentization step. $\square$

Lemma 10 (Oracle denominator LLN). *Under Assumptions 1–5 and 11, suppose*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2.$$

Then

$$V_T^{(q)} - \eta_q^2 = o_p(1).$$

Detailed solution. Conditional on $\mathcal{D}_T$, the variables $(U_t^2 - \sigma_t^2)v_{t,0}^2$ are independent and centered. Conditional sub-Gaussianity gives a uniform fourth-moment bound, hence

$$\text{Var} \left(T^{-1} \sum_{t=1}^T (U_t^2 - \sigma_t^2)v_{t,0}^2 \mid \mathcal{D}_T \right) \leq CT^{-2} \sum_{t=1}^T v_{t,0}^4.$$

Using $\max_t |v_{t,0}| = O_p(b_K)$ and $T^{-1} \sum_t v_{t,0}^2 = O_p(1)$, the right side is $O_p(b_K^2/T) = o_p(1)$. Thus

$$T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 - T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 = o_p(1),$$

and the displayed variance convergence proves the lemma. $\square$

Lemma 11 (Oracle maximum bound). *Under Assumptions 1, 3, 5, and 11,*

$$\max_{t \leq T} |U_t v_{t,0}| = o_p(\sqrt{T}).$$

Detailed solution. Conditional on $\mathcal{D}_T$, sub-Gaussian maximal inequalities give

$$\max_{t \leq T} |U_t| = O_p(\sqrt{\log T}).$$

Lemma 8 and Assumption 11 give $\max_t |v_{t,0}| = O_p(b_K)$. Therefore

$$\max_{t \leq T} |U_t v_{t,0}| = O_p(b_K \sqrt{\log T}) = o_p(\sqrt{T}),$$

by $b_K^2 \log T/T \rightarrow 0$. $\square$

Lemma 12 (Feasible true-partition replacement). *Under Assumptions 1–6 and 11, suppose*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2.$$

Then

$$T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + o_p(1), \quad (5.1)$$

$$T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0)^2 = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 + o_p(1), \quad (5.2)$$

$$\max_{t \leq T} |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}). \quad (5.3)$$

Detailed solution. Let $\mathbf{Q}_0 = \mathbf{Q}_{K, \Lambda_0}$, $\mathbf{\Pi}_0 = \mathbf{I}_T - \mathbf{M}_{K, \Lambda_0}$, and $\mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}$. The sieve representation gives

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_0 \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U}.$$

By Lemma 7,

$$\widehat{\mathbf{u}}(\beta_0, \Lambda_0) = \mathbf{M}_{K, \Lambda_0}(\mathbf{r}_0 + \mathbf{U}).$$

Define

$$\mathbf{e}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{r}_0, \quad \mathbf{h}_0 = \mathbf{\Pi}_0 \mathbf{U}, \quad \mathbf{d}_0 = (\mathbf{e}_0 - \mathbf{h}_0) \odot \mathbf{v}_0.$$

Then

$$\widehat{Z}_t(\beta_0, \Lambda_0) = U_t v_{t,0} + d_{t,0}.$$

Conditional on $\mathcal{D}_T$, $\mathbf{Q}'_0 \mathbf{U}$ is a sub-Gaussian vector with second moment of order Td_K . Hence

$$\|\mathbf{Q}'_0 \mathbf{U}\| = O_p(\sqrt{Td_K}).$$

The Gram bounds imply

$$\|\mathbf{h}_0\|_T = O_p(\sqrt{d_K/T}), \quad \|\mathbf{h}_0\|_\infty = O_p\left(\zeta_K \sqrt{d_K/T}\right).$$

Projection contraction and Assumption 6 give

$$\|\mathbf{e}_0\|_T = O_p(a_K), \quad \|\mathbf{e}_0\|_\infty = O_p\{a_K(1 + \zeta_K)\}.$$

Since $\|\mathbf{v}_0\|_\infty = O_p(b_K)$,

$$\|\mathbf{d}_0\|_T = O_p\left(a_K b_K + b_K \sqrt{d_K/T}\right) = o_p(1),$$

and

$$\|\mathbf{d}_0\|_\infty = O_p\left(a_K b_K^2 + \zeta_K b_K \sqrt{d_K/T}\right) = o_p(\sqrt{T}).$$

The numerator has the exact cancellation

$$T^{-1/2} \sum_t \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \mathbf{U}' \mathbf{v}_0 + T^{-1/2} \mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w} = T^{-1/2} \mathbf{U}' \mathbf{v}_0 + o_p(1).$$

For the denominator,

$$\left| T^{-1} \sum_t \widehat{Z}_t^2 - T^{-1} \sum_t U_t^2 v_{t,0}^2 \right| \leq 2 \|\mathbf{U} \odot \mathbf{v}_0\|_T \|\mathbf{d}_0\|_T + \|\mathbf{d}_0\|_T^2 = o_p(1),$$

because Lemma 10 gives $\|\mathbf{U} \odot \mathbf{v}_0\|_T = O_p(1)$. The maximum bound follows from Lemma 11 and $\|\mathbf{d}_0\|_\infty = o_p(\sqrt{T})$. $\square$

Remark 5 (How the stochastic lemmas enter the KKT geometry). The nuisance geometry has already reduced the feasible score to the oracle array $U_t v_{t,0}$ plus a negligible residual. The next three lemmas do not identify β_0 ; they localize the EL KKT. The numerator CLT gives the scale of $P_T Z$, the denominator LLN gives the $L^2(P_T)$ curvature $P_T Z^2$, and the maximum bound guarantees that $|\lambda_T| \max_t |Z_t| = o_p(1)$, so the entropy KKT is locally quadratic.

Theorem 1 (Known-partition conditional-design Wilks). *Under Assumptions 1–6, 9, and 11, suppose*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad \mathbb{P}(\eta_q \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$. Then

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2.$$

Detailed solution. Let

$$A_T = T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0), \quad B_T = T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0)^2.$$

Lemma 12, followed by Lemmas 9 and 10, gives

$$A_T \xrightarrow{\text{st}} \eta_q Z, \quad B_T \xrightarrow{p} \eta_q^2.$$

Because $\eta_q^2 \geq c_\eta$ almost surely, $B_T \geq c_\eta/2$ with probability tending to one. Lemmas 11 and 12 give

$$\max_t |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}).$$

Assumption 9 supplies the convex-hull condition. Lemma 5 localizes the EL multiplier, and Lemma 6 gives the local quadratic form

$$\ell_T(\beta_0, \Lambda_0) = A_T^2/B_T + o_p(1),$$

and stable Slutsky yields $A_T^2/B_T \Rightarrow Z^2 \sim \chi_1^2$. □

Remark 6 (Proof audit for Theorem 1). The known-partition result now uses a single stochastic route. Conditional on $\mathcal{D}_T$, the residualized score weights are fixed, so the numerator CLT, denominator LLN, and maximum bound are ordinary conditional independent-array arguments. Break-date localization and order selection are irrelevant until Λ_0 is replaced by an estimated partition.

6 One-Break Estimated-Partition Bridge

In this section only, take $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\widetilde{\Lambda} = (\widetilde{k})$. Let

$$\Delta_M = \mathbf{M}_{K, \widetilde{\Lambda}} - \mathbf{M}_{K, \Lambda_0}, \quad \mathcal{M}_T = \{t : j_{\widetilde{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}.$$

Lemma 13 (Boundary-set size). *If $|\widetilde{k} - k_0| = O_p(r_T)$, then*

$$|\mathcal{M}_T| = O_p(r_T).$$

Detailed solution. Suppose first that $\widetilde{k} \geq k_0$. For $t \leq k_0$, both partitions assign observation t to the left regime; for $t > \widetilde{k}$, both assign it to the right regime. The assignments differ exactly on

$$k_0 < t \leq \widetilde{k}.$$

Thus $|\mathcal{M}_T| = \widetilde{k} - k_0$. If $\widetilde{k} < k_0$, the same argument gives the differing set $\widetilde{k} < t \leq k_0$, so $|\mathcal{M}_T| = k_0 - \widetilde{k}$. Hence, in all cases,

$$|\mathcal{M}_T| = |\widetilde{k} - k_0| = O_p(r_T).$$

□

Lemma 14 (Search-uniform projection noise). *Suppose $q_0 = 1$ and Assumptions 1, 3, 5, and 13 hold. For every fixed $C < \infty$, over the local one-break set*

$$\mathcal{N}_T(Cr_T) = \{k : |k - k_0| \leq Cr_T\},$$

$$\sup_{k \in \mathcal{N}_T(Cr_T)} \|\mathbf{\Pi}_k \mathbf{U}\|_T = O_p \left(\sqrt{\frac{K + \log(1 + r_T)}{T}} \right),$$

and

$$\sup_{k \in \mathcal{N}_T(Cr_T)} \|\mathbf{\Pi}_k \mathbf{U}\|_\infty = O_p \left(\zeta_K \sqrt{\frac{\log\{T(1 + r_T)\}}{T}} \right).$$

Detailed solution. Condition on $\mathcal{D}_T$. For fixed k , $\mathbf{\Pi}_k \mathbf{U}$ is the projection of a conditionally sub-Gaussian vector onto a subspace of rank at most $2K$. Standard chi-square/sub-Gaussian projection concentration gives

$$\|\mathbf{\Pi}_k \mathbf{U}\|^2 = O_p(K + x)$$

with conditional tail probability bounded by e^{-cx} . A union bound over $O(1 + r_T)$ local candidates yields the displayed T -norm bound. For the sup norm, the leverage bound gives $(\mathbf{\Pi}_k)_{tt} \leq C\zeta_K^2/T$ uniformly over t and local k . Conditional sub-Gaussian tails for each coordinate, followed by a union bound over $T(1 + r_T)$ pairs (t, k) , give the displayed sup-norm bound. $\square$

Lemma 15 (Nuisance part of partition perturbation). *Suppose $q_0 = 1$, Assumptions 3, 5, 6, and 12 hold, and $|\tilde{k} - k_0| = O_p(r_T)$. Then*

$$T^{-1/2} \boldsymbol{\mu}' \Delta_M \mathbf{w} = O_p \left(\frac{b_K r_T}{\sqrt{T}} \right) + o_p(1), \quad b_K = 1 + \zeta_K.$$

In particular, this term is $o_p(1)$ if $b_K r_T / \sqrt{T} \rightarrow 0$.

Detailed solution. It suffices to prove a uniform deterministic bound for all $|k - k_0| \leq Cr_T$, then evaluate it at $k = \tilde{k}$ on the localization event. Consider first $k = k_0 + h$, $h \geq 0$, and let $H = \{k_0 + 1, \dots, k_0 + h\}$. Let $\mathbf{c}_0$ and $\mathbf{c}_1$ be the true-regime sieve coefficients. If the candidate partition uses $\mathbf{c}_0$ on its left block and $\mathbf{c}_1$ on its right block, then

$$\boldsymbol{\mu} = \mathbf{Q}_{K,k} \mathbf{c}^{(k)} + \mathbf{d}_H + \mathbf{r}_0,$$

where $\mathbf{d}_H$ is supported on H . For $t \in H$,

$$d_{H,t} = \mathbf{P}_K(W_{t-1})'(\mathbf{c}_1 - \mathbf{c}_0) = m_1(W_{t-1}) - m_0(W_{t-1}) - \{r_{1,K,t} - r_{0,K,t}\}.$$

Assumptions 3 and 6 imply $\|\mathbf{d}_H\|_\infty = O_p(1)$. Since $\mathbf{Q}'_{K,k} \mathbf{M}_{K,k} = \mathbf{0}$,

$$\boldsymbol{\mu}' \mathbf{M}_{K,k} \mathbf{w} = \mathbf{d}_H' \mathbf{M}_{K,k} \mathbf{w} + \mathbf{r}_0' \mathbf{M}_{K,k} \mathbf{w}.$$

At the true partition,

$$\boldsymbol{\mu}' \mathbf{M}_{K,k_0} \mathbf{w} = \mathbf{r}_0' \mathbf{M}_{K,k_0} \mathbf{w}.$$

Therefore, using Lemma 8 uniformly over local candidates,

$$\begin{aligned} T^{-1/2}|\boldsymbol{\mu}'(\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0})\mathbf{w}| &\leq T^{-1/2}\|\mathbf{d}_H\|_1\|\mathbf{M}_{K,k}\mathbf{w}\|_\infty \\ &\quad + T^{-1/2}|\mathbf{r}'_0(\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0})\mathbf{w}| \\ &= O_p\left(\frac{b_K h}{\sqrt{T}}\right) + o_p(1), \end{aligned}$$

where the final term is Assumption 12. The case $k < k_0$ is identical after exchanging the left and right regimes. Since $|\tilde{k} - k_0| = O_p(r_T)$, the lemma follows.

The sufficient primitive rate in (3.2) follows from the same local coefficient update. Write $\mathbf{v}_k = \mathbf{M}_{K,k}\mathbf{w}$. On the boundary set H , $|v_{t,k} - v_{t,k_0}| = O_p(b_K)$. Off the boundary, the candidate and true fitted-weight coefficients differ by $O_p(h\zeta_K b_K/T)$, and each row has norm at most ζ_K . Hence

$$\|\mathbf{v}_k - \mathbf{v}_{k_0}\|_1 = O_p\{hb_K(1 + \zeta_K^2)\}.$$

Consequently,

$$T^{-1/2}|\mathbf{r}'_0(\mathbf{v}_k - \mathbf{v}_{k_0})| \leq T^{-1/2}\|\mathbf{r}_0\|_\infty\|\mathbf{v}_k - \mathbf{v}_{k_0}\|_1 = O_p\left(\frac{a_K b_K(1 + \zeta_K^2)h}{\sqrt{T}}\right),$$

which is uniformly $o_p(1)$ for $h \leq Cr_T$ under (3.2). $\square$

Lemma 16 (Stochastic part of partition perturbation). *Suppose $q_0 = 1$, Assumptions 1, 3, 5, and 13 hold, $|\tilde{k} - k_0| = O_p(r_T)$, and $b_K r_T/\sqrt{T} \rightarrow 0$. Then*

$$T^{-1/2}\mathbf{U}'\Delta_M\mathbf{w} = O_p\left(\zeta_K\sqrt{\frac{r_T\{K + \log(1 + r_T)\}}{T}}\right) + o_p(1).$$

In particular, this term is $o_p(1)$ if $\zeta_K\sqrt{r_T\{K + \log(1 + r_T)\}}/T \rightarrow 0$.

Detailed solution. Work uniformly over deterministic local candidates $|k - k_0| \leq Cr_T$, then evaluate at $k = \tilde{k}$ on the localization event. Take $k = k_0 + h > k_0$ and define

$$A = \{1, \dots, k_0\}, \quad H = \{k_0 + 1, \dots, k_0 + h\}, \quad B = \{k_0 + h + 1, \dots, T\}, \quad R = H \cup B.$$

For a segment J , put

$$\mathbf{G}_J = \mathbf{P}'_J \mathbf{P}_J, \quad \hat{\boldsymbol{\gamma}}_J = \mathbf{G}_J^{-1} \mathbf{P}'_J \mathbf{w}_J.$$

Assumption 13 gives $\|\mathbf{G}_J^{-1}\| = O_p(T^{-1})$ for all long local segments, and bounded weights imply $\|\hat{\boldsymbol{\gamma}}_J\| = O_p(1)$. The exact update identity

$$\hat{\boldsymbol{\gamma}}_{A \cup H} - \hat{\boldsymbol{\gamma}}_A = \mathbf{G}_{A \cup H}^{-1} \mathbf{P}'_H (\mathbf{w}_H - \mathbf{P}_H \hat{\boldsymbol{\gamma}}_A)$$

therefore yields

$$\|\hat{\boldsymbol{\gamma}}_{A \cup H} - \hat{\boldsymbol{\gamma}}_A\| = O_p\left(\frac{h\zeta_K b_K}{T}\right).$$

The same calculation after removing H from the true right segment gives

$$\|\hat{\gamma}_B - \hat{\gamma}_R\| = O_p\left(\frac{h\zeta_K b_K}{T}\right).$$

The residualized weights are segmentwise

$$\begin{aligned} \mathbf{v}_{k_0} &= (\mathbf{w}_A - \mathbf{P}_A \hat{\gamma}_A, \mathbf{w}_R - \mathbf{P}_R \hat{\gamma}_R), \\ \mathbf{v}_k &= (\mathbf{w}_{A \cup H} - \mathbf{P}_{A \cup H} \hat{\gamma}_{A \cup H}, \mathbf{w}_B - \mathbf{P}_B \hat{\gamma}_B). \end{aligned}$$

Cancelling the raw $\mathbf{w}_H$ terms gives the exact decomposition

$$\begin{aligned} \mathbf{U}'(\mathbf{v}_k - \mathbf{v}_{k_0}) &= (\mathbf{P}'_A \mathbf{U}_A)'(\hat{\gamma}_A - \hat{\gamma}_{A \cup H}) \\ &\quad + (\mathbf{P}'_B \mathbf{U}_B)'(\hat{\gamma}_R - \hat{\gamma}_B) \\ &\quad + (\mathbf{P}'_H \mathbf{U}_H)'(\hat{\gamma}_R - \hat{\gamma}_{A \cup H}). \end{aligned} \tag{6.1}$$

Conditional on $\mathcal{D}_T$, sub-Gaussian concentration and the Gram bounds give

$$\|\mathbf{P}'_A \mathbf{U}_A\| = O_p(\sqrt{TK}), \quad \|\mathbf{P}'_B \mathbf{U}_B\| = O_p(\sqrt{TK}).$$

For the local moving block, a union bound over $h \leq Cr_T$ gives

$$\sup_{h \leq Cr_T} \|\mathbf{P}'_H \mathbf{U}_H\| = O_p\{\zeta_K \sqrt{r_T \log(1 + r_T)}\}.$$

The coefficient difference in the last line of (6.1) is $O_p(1)$. Combining the displays, uniformly for $h \leq Cr_T$,

$$T^{-1/2} |\mathbf{U}'(\mathbf{v}_k - \mathbf{v}_{k_0})| = O_p\left(\frac{h\zeta_K b_K \sqrt{K}}{T}\right) + O_p\left(\zeta_K \sqrt{\frac{r_T \log(1 + r_T)}{T}}\right).$$

The first order is negligible under $b_K r_T / \sqrt{T} \rightarrow 0$, and the second order is exactly the logarithmic local-search term. Taking $h = O_p(r_T)$ proves the displayed result. The case $k < k_0$ is the same with left and right interchanged. $\square$

Lemma 17 (Denominator perturbation). *Suppose $q_0 = 1$, Assumptions 1, 2, 3, 5, 6, 11, 12, and 13 hold. Suppose also*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad |\tilde{k} - k_0| = O_p(r_T),$$

and

$$\frac{b_K r_T}{\sqrt{T}} \rightarrow 0, \quad b_K \sqrt{\frac{K + \log(1 + r_T)}{T}} \rightarrow 0, \quad \frac{\zeta_K^2 K r_T}{T} \rightarrow 0.$$

Then

$$V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1) = o_p(1).$$

Detailed solution. Again work uniformly over $|k - k_0| \leq Cr_T$. Put $\mathbf{v}_k = \mathbf{M}_{K,k}\mathbf{w}$ and $V_{U,T}(k) = T^{-1} \sum_t U_t^2 v_{t,k}^2$. As in Lemma 15, for a local candidate partition,

$$\boldsymbol{\mu} = \mathbf{Q}_{K,k}\mathbf{c}^{(k)} + \mathbf{d}_H + \mathbf{r}_0,$$

where $\mathbf{d}_H$ is supported on the boundary set and $\|\mathbf{d}_H\|_\infty = O_p(1)$. Therefore

$$\hat{\mathbf{u}}(\beta_0, k) = \mathbf{U} - \boldsymbol{\Pi}_k \mathbf{U} + \mathbf{M}_{K,k} \mathbf{r}_0 + \mathbf{M}_{K,k} \mathbf{d}_H, \quad \boldsymbol{\Pi}_k = \mathbf{I}_T - \mathbf{M}_{K,k},$$

and

$$\hat{\mathbf{Z}}(\beta_0, k) = \mathbf{U} \odot \mathbf{v}_k + \boldsymbol{\rho}_k,$$

with

$$\|\boldsymbol{\rho}_k\|_T \leq \|\mathbf{v}_k\|_\infty \{\|\boldsymbol{\Pi}_k \mathbf{U}\|_T + \|\mathbf{M}_{K,k} \mathbf{r}_0\|_T + \|\mathbf{M}_{K,k} \mathbf{d}_H\|_T\}.$$

Lemma 14 gives, uniformly over the local full-rank set,

$$\|\boldsymbol{\Pi}_k \mathbf{U}\|_T = O_p \left(\sqrt{\frac{K + \log(1 + r_T)}{T}} \right).$$

Projection contraction gives

$$\|\mathbf{M}_{K,k} \mathbf{r}_0\|_T = O_p(a_K), \quad \|\mathbf{M}_{K,k} \mathbf{d}_H\|_T = O_p(\sqrt{h/T}).$$

Since $\|\mathbf{v}_k\|_\infty = O_p(b_K)$, Assumptions 6 and 11, together with $b_K r_T / \sqrt{T} \rightarrow 0$, imply

$$\sup_{|k - k_0| \leq Cr_T} \|\boldsymbol{\rho}_k\|_T = o_p(1). \quad (6.2)$$

It remains to compare the oracle denominators. The coefficient-update bounds from Lemma 16 imply, for $h = |k - k_0|$,

$$|v_{t,k} - v_{t,k_0}| \leq \begin{cases} Cb_K, & t \in \mathcal{M}_T, \\ Ch\zeta_K^2 b_K / T, & t \notin \mathcal{M}_T. \end{cases}$$

Also $\max_t (|v_{t,k}| + |v_{t,k_0}|) = O_p(b_K)$. For a boundary interval of length h , conditional sub-Gaussian concentration and a dyadic union bound give

$$\sup_{1 \leq h \leq Cr_T} \frac{1}{h} \sum_{t \in H_h} U_t^2 = O_p(1),$$

and $T^{-1} \sum_t U_t^2 = O_p(1)$. Hence, uniformly over local candidates,

$$\begin{aligned} |V_{U,T}(k) - V_{U,T}(k_0)| &\leq T^{-1} \sum_t U_t^2 |v_{t,k} - v_{t,k_0}| (|v_{t,k}| + |v_{t,k_0}|) \\ &= O_p \left(\frac{b_K^2 h}{T} \right) + O_p \left(\frac{h \zeta_K^2 b_K^2}{T} \right). \end{aligned}$$

Assumption 13 gives $b_K^2 = O(K)$, so the second term is $O_p(\zeta_K^2 Kh/T)$. Thus

$$|V_{U,T}(k) - V_{U,T}(k_0)| = O_p\left(\frac{b_K^2 h}{T} + \frac{\zeta_K^2 Kh}{T}\right).$$

By Lemma 10, $V_{U,T}(k_0) = O_p(1)$; hence $\|\mathbf{U} \odot \mathbf{v}_k\|_T = O_p(1)$ uniformly locally. The square-difference inequality and (6.2) imply

$$V_T(k) - V_{U,T}(k) = o_p(1)$$

uniformly locally, and the same is true at k_0 . Taking $h = |\tilde{k} - k_0| = O_p(r_T)$ gives the displayed bound. $\square$

Theorem 2 (One-break estimated-partition score equivalence). *Suppose $q_0 = 1$, Assumptions 1–8, 11, 12, and 13 hold, and*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2.$$

Assume $r_T \geq 1$ and the bridge rates

$$\zeta_K \sqrt{\frac{r_T \{K + \log(1 + r_T)\}}{T}} \rightarrow 0, \quad \frac{b_K r_T}{\sqrt{T}} \rightarrow 0, \quad \frac{\zeta_K^2 K r_T}{T} \rightarrow 0.$$

Then

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = o_p(1).$$

Detailed solution. Assumption 8 gives $|\tilde{k} - k_0| = O_p(r_T)$. For every one-break partition Λ , symmetry and idempotence imply

$$S_T(\Lambda) = T^{-1/2}(\mathbf{U} + \boldsymbol{\mu})' \mathbf{M}_{K,\Lambda} \mathbf{w}.$$

Therefore

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = T^{-1/2} \boldsymbol{\mu}' \Delta_M \mathbf{w} + T^{-1/2} \mathbf{U}' \Delta_M \mathbf{w}.$$

Lemma 15 bounds the nuisance term by

$$O_p\left(\frac{b_K r_T}{\sqrt{T}}\right) + o_p(1) = o_p(1),$$

and Lemma 16 bounds the stochastic term by

$$O_p\left(\zeta_K \sqrt{\frac{r_T \{K + \log(1 + r_T)\}}{T}}\right) + o_p(1) = o_p(1).$$

This proves the numerator equivalence.

For the denominator, Lemma 17 gives

$$V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1).$$

The third bridge rate makes the second term $o_p(1)$. Since $r_T \geq 1$,

$$\frac{b_K^2 r_T}{T} = \frac{1}{r_T} \left(\frac{b_K r_T}{\sqrt{T}} \right)^2 = o(1),$$

so the denominator difference is $o_p(1)$. $\square$

Remark 7 (Proof audit for Theorem 2). Theorem 2 is only a bridge from the true one-break partition to a localized estimated break. Lemma 13 limits the number of observations whose regime assignment can change. Lemma 15 handles the deterministic nuisance component and makes the local sieve-bias condition explicit. Lemma 16 handles the stochastic component through conditional sub-Gaussian concentration and segmentwise coefficient updates. Lemma 17 proves that the empirical second moment is stable after the same partition perturbation. No convex-hull condition and no empirical-likelihood expansion is used in this theorem; those enter only in the next Wilks theorem. The result would fail without localization, local full-rank Gram regularity, and the three bridge rates listed in the theorem statement.

Theorem 3 (One-break estimated-partition Wilks). *Under the assumptions of Theorem 1, the convex-hull condition for the estimated-partition score array, and Theorem 2,*

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2.$$

Detailed solution. Let $A_T(\Lambda) = S_T(\Lambda)$ and $B_T(\Lambda) = V_T(\Lambda)$. Theorem 2 gives

$$A_T(\tilde{\Lambda}) = A_T(\Lambda_0) + o_p(1), \quad B_T(\tilde{\Lambda}) = B_T(\Lambda_0) + o_p(1).$$

The known-partition proof gives

$$A_T(\Lambda_0) \xrightarrow{\text{st}} \eta_q Z, \quad B_T(\Lambda_0) \xrightarrow{p} \eta_q^2.$$

We must also verify the maximum condition; numerator and denominator equivalence alone would not imply it. The local representation in the proof of Lemma 17 gives

$$\hat{\mathbf{Z}}(\beta_0, \tilde{\Lambda}) = \mathbf{U} \odot \mathbf{v}_{\tilde{k}} + \boldsymbol{\rho}_{\tilde{k}}.$$

The envelope rate implies $\max_t |U_t v_{t,\tilde{k}}| = o_p(\sqrt{T})$. For the remainder, the uniform projection calculation gives

$$\|\boldsymbol{\Pi}_{\tilde{k}} \mathbf{U}\|_\infty = O_p \left(\zeta_K \sqrt{\frac{\log\{T(1+r_T)\}}{T}} \right),$$

while the sieve calculation gives

$$\|\mathbf{M}_{K,\tilde{k}} \mathbf{r}_0\|_\infty = O_p\{a_K(1 + \zeta_K)\}.$$

Finally, because $\mathbf{d}_H$ has bounded entries and support of size $O_p(r_T)$,

$$\|\mathbf{M}_{K,\tilde{k}} \mathbf{d}_H\|_\infty = O_p \left(1 + \frac{r_T \zeta_K^2}{T} \right) = O_p(1).$$

Multiplication by $\|\mathbf{v}_{\tilde{k}}\|_\infty = O_p(b_K)$, followed by the sieve, bridge, and envelope rates, gives $\|\boldsymbol{\rho}_{\tilde{k}}\|_\infty = o_p(\sqrt{T})$. Hence

$$\max_t |\widehat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}).$$

The estimated-partition convex-hull condition and Lemma 6 now yield

$$\ell_T(\beta_0, \tilde{\Lambda}) = \frac{A_T(\tilde{\Lambda})^2}{B_T(\tilde{\Lambda})} + o_p(1).$$

Substitution of the two equivalences reduces this ratio to the true-partition ratio plus $o_p(1)$, and Theorem 1 gives the χ_1^2 limit. $\square$

7 Conditional-Design Profile-Sieve Segmentation

This section is the main rebuild of the Bai–Perron module. The break search is no longer imported as a broad high-level assumption. Instead, the stochastic part is controlled by conditional-on-design sub-Gaussian concentration and the identifying part is isolated as a deterministic profile-loss condition.

At the null value β_0 , define

$$y_t(\beta_0) = Y_t - \beta_0 X_{t-1}.$$

Then the nuisance-break model and sieve approximation imply

$$y_t(\beta_0) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_{j,K} + U_t + r_{j,K,t}, \quad k_{j,0} < t \leq k_{j+1,0}.$$

Thus, apart from approximation error, the break search is a segmented least-squares problem with regime-specific sieve coefficient vectors.

Assumption 16 (Conditional-design profile-sieve segmentation). *For the main estimated-partition theory, $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\bar{q}$ is fixed. Let $\boldsymbol{\mu}_K = \mathbf{Q}_{K,\Lambda_0} \mathbf{c}_0$ be the sieve part of the true conditional mean and define*

$$D_T(\Lambda) = \|\mathbf{M}_{K,\Lambda} \boldsymbol{\mu}_K\|^2.$$

The following conditions hold.

1. **Admissible partitions.** All searches are over $\mathcal{L}_q(\epsilon)$, the true break obeys the stricter spacing $\epsilon_0 > \epsilon$, and $(\bar{q} + 1)\epsilon < 1$.
2. **Global segment conditioning.** The global Gram condition in Assumption 5 holds for every admissible segment used in the search.
3. **Deterministic profile identification.** For every admissible one-break candidate,

$$D_T(k) \geq c_\Delta |k - k_0| \Delta_T^2.$$

For the underfitted zero-break model,

$$\inf_{\Lambda \in \mathcal{L}_0(\epsilon)} D_T(\Lambda) \geq c_\Delta T \Delta_T^2.$$

4. **Uniform conditional RSS expansion.** With

$$R_T = K + \log T + Ta_K^2, \quad r_T = R_T/\Delta_T^2,$$

conditional sub-Gaussian concentration gives, uniformly over the admissible one-break search,

$$RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0) \geq \frac{1}{2}D_T(k) - O_p(R_T).$$

It also gives the order-selection comparisons

$$RSS_K\{\beta_0, \hat{\Lambda}_0(\beta_0)\} - RSS_K(\beta_0, k_0) \geq cT\Delta_T^2 - O_p(R_T),$$

and, uniformly over $1 < q \leq \bar{q}$,

$$RSS_K(\beta_0, k_0) - RSS_K\{\beta_0, \hat{\Lambda}_q(\beta_0)\} = O_p(R_T).$$

The rates satisfy

$$R_T = o(T\Delta_T^2), \quad r_T = o(\sqrt{T}).$$

Proposition 2 (Primitive verification of the profile-RSS expansion). *Suppose $q_0 = 1$, $\bar{q}$ is fixed, and Assumptions 1, 5, 6, and 14 hold. Then the uniform RSS expansion and the underfit/overfit order comparisons in Assumption 16 hold with $R_T = K + \log T + Ta_K^2$.*

Detailed solution. Condition on $\mathcal{D}_T$ and work on the Gram and sieve-approximation events whose probability tends to one. For any admissible partition Λ , write $\mathbf{y}(\beta_0) = \boldsymbol{\mu}_K + \mathbf{r}_0 + \mathbf{U}$. Since $\mathbf{M}_{K, \Lambda_0} \boldsymbol{\mu}_K = \mathbf{0}$,

$$RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) = D_T(\Lambda) + 2\mathbf{U}'\mathbf{M}_{K, \Lambda}\boldsymbol{\mu}_K + \mathbf{U}'(\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0})\mathbf{U} + \mathcal{E}_{r, T}(\Lambda),$$

where $\mathcal{E}_{r, T}(\Lambda)$ collects every term containing $\mathbf{r}_0$. The admissible search over $0 \leq q \leq \bar{q}$ has polynomial cardinality, so its logarithm is $O(\log T)$.

For fixed Λ , conditional sub-Gaussian tails give

$$2|\mathbf{U}'\mathbf{M}_{K, \Lambda}\boldsymbol{\mu}_K| \leq \frac{1}{4}D_T(\Lambda) + O_p(\log T)$$

uniformly over the search after a union bound, because the conditional variance proxy is proportional to $D_T(\Lambda)$. Since $\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0} = \mathbf{\Pi}_{K, \Lambda_0} - \mathbf{\Pi}_{K, \Lambda}$, it is a difference of two fitted-space projections whose ranks are at most $(\bar{q} + 1)K$. Hence the difference has rank $O(K)$ and operator norm bounded by a constant. Hanson–Wright concentration gives

$$\sup_{\Lambda} |\mathbf{U}'(\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0})\mathbf{U} - \mathbb{E}\{\mathbf{U}'(\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0})\mathbf{U} \mid \mathcal{D}_T\}| = O_p(K + \log T),$$

and the conditional expectation is $O(K)$ by the bounded conditional variances and the same rank bound.

Finally, projection contraction and $\|\mathbf{r}_0\|^2 = O_p(Ta_K^2)$ imply, uniformly over the search,

$$|\mathcal{E}_{r, T}(\Lambda)| \leq \frac{1}{4}D_T(\Lambda) + O_p(Ta_K^2 + K + \log T).$$

The mixed term $\mathbf{r}'_0(\mathbf{M}_{K,\Lambda} - \mathbf{M}_{K,\Lambda_0})\mathbf{U}$ is absorbed by $2ab \leq a^2 + b^2$ and the preceding projection-noise bound. Combining the three displays yields

$$RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) \geq \frac{1}{2}D_T(\Lambda) - O_p(R_T)$$

uniformly over the admissible search.

For one-break candidates, the deterministic identification inequality in Assumption 14 gives the displayed one-break RSS expansion and the localization rate $r_T = R_T/\Delta_T^2$. For the zero-break model, the underfit identification bound gives

$$RSS_K\{\beta_0, \hat{\Lambda}_0(\beta_0)\} - RSS_K(\beta_0, k_0) \geq cT\Delta_T^2 - O_p(R_T).$$

For $q > 1$, the same uniform lower bound and nonnegativity of D_T give

$$RSS_K(\beta_0, k_0) - RSS_K\{\beta_0, \hat{\Lambda}_q(\beta_0)\} = O_p(R_T),$$

because the right side is the largest possible stochastic overfit gain relative to the true one-break partition. These are exactly the order-selection comparisons in Assumption 16. $\square$

Proposition 3 (Profile-sieve reduction to segmented least squares). *At β_0 , the fixed-order nuisance-break estimator is the least-squares segmentation estimator in the generated regression*

$$y_t(\beta_0) = \mathbf{P}_K(W_{t-1})'\mathbf{c}_j + \text{error}, \quad t \in I_j(\Lambda).$$

Detailed solution. For a fixed candidate partition Λ , minimizing the null-imposed segment cost over the nuisance coefficients gives

$$RSS_K(\beta_0, \Lambda) = \sum_{j=0}^q \min_{\mathbf{c}_j \in \mathbb{R}^K} \sum_{t=k_j+1}^{k_{j+1}} \{y_t(\beta_0) - \mathbf{P}_K(W_{t-1})'\mathbf{c}_j\}^2.$$

This is the segmented least-squares criterion for a linear regression whose coefficient vector may change across regimes. The block matrix $\mathbf{Q}_{K,\Lambda}$ is the stacked design matrix and $\mathbf{M}_{K,\Lambda}$ is its residual-maker. The nonstandard work is exactly what Assumption 16 records: growing K , approximation error, uniform segment conditioning, and conditional concentration over the finite admissible search. $\square$

Lemma 18 (One-break profile-RSS exclusion). *Under Assumption 16, for every fixed M , define*

$$\mathcal{B}_T(M) = \{k \in \mathcal{L}_1(\epsilon) : |k - k_0| \geq Mr_T\}.$$

Then

$$\lim_{M \rightarrow \infty} \liminf_{T \rightarrow \infty} \mathbb{P} \left\{ \inf_{k \in \mathcal{B}_T(M)} [RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0)] > 0 \right\} = 1.$$

Detailed solution. For $k \in \mathcal{B}_T(M)$, deterministic identification and the uniform RSS expansion give

$$RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0) \geq \frac{1}{2}c_\Delta Mr_T \Delta_T^2 - O_p(R_T) = \left(\frac{1}{2}c_\Delta M + O_p(1) \right) R_T.$$

Taking $T \rightarrow \infty$ and then $M \rightarrow \infty$ proves the exclusion event. $\square$

Theorem 4 (One-break profile-sieve localization). *Under Assumption 16,*

$$|\widehat{k}_1(\beta_0) - k_0| = O_p(r_T), \quad r_T = R_T/\Delta_T^2 = o(\sqrt{T}).$$

Detailed solution. The one-break estimator minimizes $RSS_K(\beta_0, k)$ over $\mathcal{L}_1(\epsilon)$, and k_0 is admissible. On the exclusion event in Lemma 18, no candidate with $|k - k_0| \geq Mr_T$ can minimize the criterion. Hence

$$\mathbb{P}\{|\widehat{k}_1(\beta_0) - k_0| \geq Mr_T\} \leq \mathbb{P}\left\{\inf_{k \in \mathcal{B}_T(M)} [RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0)] \leq 0\right\},$$

and the right side vanishes after taking $T \rightarrow \infty$ and then $M \rightarrow \infty$. $\square$

Assumption 17 (Concrete unknown-order penalty). *Use*

$$\text{pen}_T(q, K) = q\kappa_T R_T,$$

where

$$\kappa_T \rightarrow \infty, \quad \kappa_T R_T = o(T\Delta_T^2).$$

For instance, $\kappa_T = \log \log(T + e^e)$ is valid whenever $R_T \log \log T = o(T\Delta_T^2)$.

Theorem 5 (Unknown-order consistency, one-break truth). *Under Assumptions 16 and 17, with $q_0 = 1$,*

$$\mathbb{P}\{\widehat{q}(\beta_0) = 1\} \rightarrow 1.$$

Detailed solution. Write $RSS_q^* = RSS_K\{\beta_0, \widehat{\Lambda}_q(\beta_0)\}$. For the underfitted order $q = 0$, Assumption 16 gives

$$RSS_0^* - RSS_1^* \geq cT\Delta_T^2 - O_p(R_T).$$

The penalty advantage of underfitting is only $\kappa_T R_T = o(T\Delta_T^2)$, so the penalized zero-break objective is larger with probability tending to one. For every $q > 1$, the overfit RSS gain is $O_p(R_T)$, while the penalty increase is $(q - 1)\kappa_T R_T$. Since $\kappa_T \rightarrow \infty$ and $\bar{q}$ is fixed, all overfitted objectives are larger with probability tending to one. A finite union bound over $q \neq 1$ proves the claim. $\square$

Corollary 1 (Jointly feasible rates). *Suppose $\zeta_K^2 \asymp K$, $a_K \asymp c_K \asymp K^{-s/d}$, $\Delta_T \geq c_\Delta > 0$, and choose*

$$K \asymp T^{d/(2s+d)}.$$

If $s > d$, then

$$R_T = K + \log T + Ta_K^2 \asymp K, \quad r_T = R_T/\Delta_T^2 \asymp K,$$

and all main bridge, local-bias, envelope, convex-hull transfer, and penalty rates hold with, for example,

$$\text{pen}_T(q, K) = qK \log \log(T + e^e).$$

Detailed solution. With $K \asymp T^{d/(2s+d)}$ and $a_K \asymp K^{-s/d}$, $Ta_K^2 \asymp K$, so $R_T \asymp K$ and $r_T \asymp K$. The largest bridge restriction is

$$\frac{K^3 \log K}{T} \rightarrow 0,$$

up to logarithmic factors, and this follows from $s > d$. The local-bias rate is

$$\frac{a_K K^{5/2}}{\sqrt{T}} \rightarrow 0,$$

which is also implied by $s > d$. The score-bias orthogonality route requires

$$\sqrt{T} a_K c_K \rightarrow 0,$$

which already holds under $s > d/2$ and therefore under $s > d$. The convex-hull transfer terms reduce to

$$K^{1-s/d} \rightarrow 0, \quad K \sqrt{\frac{\log T}{T}} \rightarrow 0, \quad \frac{K^{5/2}}{T} \rightarrow 0,$$

again under $s > d$. Finally, $K \log \log T = o(T\Delta_T^2)$, so the displayed penalty satisfies Assumption 17. $\square$

Lemma 19 (High-probability event transfer). *Let X_T and Y_T be real statistics, and let E_T be events such that $\mathbb{P}(E_T) \rightarrow 1$. If $X_T \Rightarrow X$ and $Y_T = X_T$ on E_T , then $Y_T \Rightarrow X$.*

Detailed solution. For every bounded continuous function f ,

$$|\mathbb{E}f(Y_T) - \mathbb{E}f(X_T)| \leq 2\|f\|_\infty \mathbb{P}(E_T^c) \rightarrow 0.$$

Since $\mathbb{E}f(X_T) \rightarrow \mathbb{E}f(X)$, the same convergence holds for $\mathbb{E}f(Y_T)$, which is weak convergence of Y_T to X . $\square$

Corollary 2 (Unknown-order Wilks, one-break truth). *Suppose $q_0 = 1$, Theorem 5 holds, the fixed-order estimated-partition Wilks theorem 3 holds for $\tilde{\Lambda} = \hat{\Lambda}_1(\beta_0)$, and the same measurable tie-breaking convention is used for fixed-order and unknown-order selection. Then*

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Detailed solution. Let $E_T = \{\hat{q}(\beta_0) = 1\}$. By Theorem 5, $\mathbb{P}(E_T) \rightarrow 1$. On E_T , unknown-order selection uses the one-break order and the same deterministic tie-breaking rule as the fixed-order estimator, so

$$\hat{\Lambda}(\beta_0) = \hat{\Lambda}_1(\beta_0) = \tilde{\Lambda}.$$

Therefore

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} = \ell_T(\beta_0, \tilde{\Lambda})$$

on E_T . The fixed-order estimated-partition Wilks theorem gives $\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2$, and Lemma 19 transfers this limit to the unknown-order statistic. $\square$

Remark 8 (Proof audit for Corollary 2). The corollary uses only order consistency and event transfer, and it is stated only for the one-break truth covered by Theorem 3. It does not introduce a new empirical-likelihood expansion. The deterministic tie-breaking convention is essential: without it, the selected partition on E_T need not be literally the same measurable object as $\hat{\Lambda}_1(\beta_0)$.

Theorem 6 (Master one-break conditional-design Wilks). *Suppose $q_0 = 1$, $\bar{q}$ is fixed, and Assumptions 1–6, 10, 11, 12, 13, 14, and 17 hold. Suppose also*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta^2, \quad \mathbb{P}(\eta \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$, and the bridge rates in Theorem 2 hold. Then, for the fixed one-break profile estimator $\tilde{\Lambda} = \hat{\Lambda}_1(\beta_0)$,

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2.$$

Moreover, for the unknown-order selector over $0 \leq q \leq \bar{q}$ with the same deterministic tie-breaking rule,

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Detailed solution. Proposition 2 verifies Assumption 16. Theorem 4 then verifies the localization input in Assumption 8 for $\tilde{\Lambda} = \hat{\Lambda}_1(\beta_0)$. Theorem 2 then transfers the first two empirical-score moments from Λ_0 to $\tilde{\Lambda}$. Lemma 3 verifies the convex-hull condition for both the true and estimated one-break scores under Assumption 10 and the displayed transfer rate. The known-partition Wilks theorem and the estimated-partition bridge therefore give $\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2$.

For unknown order, Theorem 5 gives $\mathbb{P}\{\hat{q}(\beta_0) = 1\} \rightarrow 1$. On that event the unknown-order partition equals $\tilde{\Lambda}$ by the common tie-breaking convention, so Lemma 19 transfers the fixed-order Wilks limit to the unknown-order statistic. $\square$

8 Local Alternatives

Assumption 18 (Canonical local alternative and drift). *Let*

$$\beta_T = \beta_0 + \frac{h}{\sqrt{T}}, \quad h \in \mathbb{R}$$

with deterministic h . Define

$$\Gamma_T = T^{-1} \mathbf{X}'_{lag} \mathbf{M}_{K, \Lambda_0} \mathbf{w} = T^{-1} \sum_{t=1}^T (\mathbf{M}_{K, \Lambda_0} \mathbf{X}_{lag})_t v_{t,0}.$$

Under P_{β_T} , Assumption 1 and the Gram/envelope conditions continue to hold, and

$$\Gamma_T \xrightarrow{p} \Gamma, \quad \eta_T^2 := T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta^2, \quad 0 < \eta^2 < \infty,$$

where Γ and η are deterministic. Also

$$T^{-1} \sum_{t=1}^T \{(\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}\}^2 = O_p(1), \quad \max_t |(\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}| = o_p(T).$$

Put

$$g_{t,T} = \frac{h}{\sqrt{T}} (\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}, \quad \delta_h = \frac{h\Gamma}{\eta}.$$

Assumption 19 (Local drift stability under break perturbation). *For every fixed $C < \infty$, uniformly over $|k - k_0| \leq Cr_T$,*

$$T^{-1/2} |\mathbf{X}'_{lag} \{\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0}\} \mathbf{w}| = o_p(1),$$

where $q_0 = 1$. With

$$g_t(k) = \frac{h}{\sqrt{T}} (\mathbf{M}_{K,k} \mathbf{X}_{lag})_t (\mathbf{M}_{K,k} \mathbf{w})_t,$$

we also have

$$\sup_{|k-k_0| \leq Cr_T} T^{-1} \sum_{t=1}^T \{g_t(k) - g_t(k_0)\}^2 = o_p(1), \quad \sup_{|k-k_0| \leq Cr_T} \max_t |g_t(k) - g_t(k_0)| = o_p(\sqrt{T}).$$

Finally, under P_{β_T} , the profile-RSS expansion in Assumption 16 remains valid with the same R_T .

Proposition 4 (Local-law bridge and order stability). *Suppose $q_0 = 1$, Assumptions 18 and 19 hold, and the null-law one-break localization, bridge, and unknown-order penalty rates hold. Under P_{β_T} ,*

$$|\tilde{k} - k_0| = O_p(r_T),$$

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = o_p(1), \quad \max_t |\hat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}),$$

and, if unknown order is used,

$$P_{\beta_T} \{\hat{q}(\beta_0) = 1\} \rightarrow 1.$$

Detailed solution. Under $\beta_T = \beta_0 + h/\sqrt{T}$, the null-imposed segmentation criterion adds only the local slope-drift terms generated by $h\mathbf{X}_{lag}/\sqrt{T}$. Assumption 19 states precisely that these terms are $o_p(1)$ at the empirical-likelihood score scale and are absorbed into the same $O_p(R_T)$ RSS remainder used under the null. Hence the proof of Theorem 4 and Theorem 5 applies under P_{β_T} . The numerator and denominator bridge proofs gain the drift array $g_t(k) - g_t(k_0)$, and the two displayed drift-stability bounds make its contribution negligible in the first two score moments and in the maximum-score condition. The remaining terms are exactly the null-law bridge terms already controlled in Theorem 2. $\square$

Lemma 20 (Local shifted-score replacement). *Suppose Assumptions 1, 6, 3, 5, and 11 continue to hold under P_{β_T} . Then*

$$T^{-1/2} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \Lambda_0) - (U_t v_{t,0} + g_{t,T})\} = o_p(1), \quad (8.1)$$

$$T^{-1} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \Lambda_0)^2 - (U_t v_{t,0} + g_{t,T})^2\} = o_p(1), \quad (8.2)$$

$$\max_t |\widehat{Z}_t(\beta_0, \Lambda_0) - (U_t v_{t,0} + g_{t,T})| = o_p(\sqrt{T}). \quad (8.3)$$

Detailed solution. Under P_{β_T} ,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_{K, \Lambda_0} \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U} + \frac{h}{\sqrt{T}} \mathbf{X}_{lag}.$$

Apply $\mathbf{M}_{K, \Lambda_0}$ and put

$$\mathbf{e}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{r}_0, \quad \mathbf{h}_0 = \mathbf{\Pi}_{K, \Lambda_0} \mathbf{U}, \quad \mathbf{x}_0^c = \mathbf{M}_{K, \Lambda_0} \mathbf{X}_{lag}.$$

Then

$$\widehat{\mathbf{u}}(\beta_0, \Lambda_0) = \mathbf{U} - \mathbf{h}_0 + \mathbf{e}_0 + \frac{h}{\sqrt{T}} \mathbf{x}_0^c.$$

Multiplication by $\mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}$ gives the exact identity

$$\widehat{Z}_t(\beta_0, \Lambda_0) = U_t v_{t,0} + g_{t,T} + e_{t,T}^{\text{feas}}, \quad e_{t,T}^{\text{feas}} = (e_{t,0} - h_{t,0}) v_{t,0}.$$

The vector $\mathbf{e}_T^{\text{feas}}$ is the same feasibility error controlled in Lemma 12; under the stated local-law continuation of the primitive assumptions,

$$T^{-1/2} \sum_t e_{t,T}^{\text{feas}} = o_p(1), \quad \|\mathbf{e}_T^{\text{feas}}\|_T = o_p(1), \quad \|\mathbf{e}_T^{\text{feas}}\|_\infty = o_p(\sqrt{T}).$$

The first and third displays of the lemma follow immediately.

For the square replacement, set $a_{t,T} = U_t v_{t,0} + g_{t,T}$. Then

$$T^{-1} \sum_t \{\widehat{Z}_t^2 - a_{t,T}^2\} = 2T^{-1} \sum_t a_{t,T} e_{t,T}^{\text{feas}} + T^{-1} \sum_t (e_{t,T}^{\text{feas}})^2.$$

Assumption 18 gives $\|\mathbf{U} \odot \mathbf{v}_0\|_T = O_p(1)$ and $\|\mathbf{g}_T\|_T = o_p(1)$, hence $\|\mathbf{a}_T\|_T = O_p(1)$. By Cauchy–Schwarz and $\|\mathbf{e}_T^{\text{feas}}\|_T = o_p(1)$, both terms on the right are $o_p(1)$. $\square$

Lemma 21 (True-partition local empirical-likelihood limit). *Under Assumption 18, Lemma 20, and the local convex-hull condition for $\widehat{Z}_t(\beta_0, \Lambda_0)$,*

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2(\delta_h^2).$$

Detailed solution. Lemma 20, the conditional-design CLT from Lemma 9 under the local law, and Assumption 18 give

$$\begin{aligned} T^{-1/2} \sum_t \widehat{Z}_t(\beta_0, \Lambda_0) &= T^{-1/2} \sum_t U_t v_{t,0} + T^{-1/2} \sum_t g_{t,T} + o_p(1) \\ &\xrightarrow{\text{st}} \eta Z + h\Gamma = \eta(Z + \delta_h). \end{aligned}$$

For the second moment,

$$T^{-1} \sum_t (U_t v_{t,0} + g_{t,T})^2 = T^{-1} \sum_t U_t^2 v_{t,0}^2 + 2T^{-1} \sum_t U_t v_{t,0} g_{t,T} + T^{-1} \sum_t g_{t,T}^2.$$

The first term converges to η^2 and the last term is $o_p(1)$. The cross term is $o_p(1)$ by Cauchy-Schwarz:

$$\left| T^{-1} \sum_t U_t v_{t,0} g_{t,T} \right| \leq \left(T^{-1} \sum_t U_t^2 v_{t,0}^2 \right)^{1/2} \left(T^{-1} \sum_t g_{t,T}^2 \right)^{1/2} = o_p(1).$$

The local replacement lemma therefore gives

$$T^{-1} \sum_t \widehat{Z}_t(\beta_0, \Lambda_0)^2 \xrightarrow{p} \eta^2.$$

The maximum condition follows from

$$\max_t |U_t v_{t,0} + g_{t,T}| = o_p(\sqrt{T})$$

and the maximum replacement bound in Lemma 20. The scalar empirical-likelihood expansion, Lemma 6, now yields

$$\ell_T(\beta_0, \Lambda_0) = \frac{\{\eta(Z + \delta_h)\}^2}{\eta^2} + o_p(1) \Rightarrow (Z + \delta_h)^2 \sim \chi_1^2(\delta_h^2).$$

□

Theorem 7 (Local noncentral chi-square). *Suppose the conditions of Lemma 21 hold, Proposition 4 holds, and the local convex-hull condition holds for $\widehat{Z}_t(\beta_0, \tilde{\Lambda})$. Then*

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2(\delta_h^2).$$

If $q_0 = 1$ and $P_{\beta_T}\{\widehat{q}(\beta_0) = 1\} \rightarrow 1$, the same limit holds for $\ell_T\{\beta_0, \widehat{\Lambda}(\beta_0)\}$.

Detailed solution. The proof of Lemma 21 gives the true-partition score limits

$$S_T(\Lambda_0) \xrightarrow{\text{st}} \eta(Z + \delta_h), \quad V_T(\Lambda_0) \xrightarrow{p} \eta^2, \quad \max_t |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}).$$

The local bridge transfers the first two limits and supplies the maximum bound for the estimated partition:

$$S_T(\tilde{\Lambda}) \xrightarrow{\text{st}} \eta(Z + \delta_h), \quad V_T(\tilde{\Lambda}) \xrightarrow{p} \eta^2, \quad \max_t |\widehat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}).$$

The local convex-hull condition and Lemma 6 give

$$\ell_T(\beta_0, \tilde{\Lambda}) = \frac{S_T(\tilde{\Lambda})^2}{V_T(\tilde{\Lambda})} + o_p(1) \Rightarrow (Z + \delta_h)^2 \sim \chi_1^2(\delta_h^2).$$

For unknown order, let $E_T = \{\hat{q}(\beta_0) = 1\}$. On E_T , $\hat{\Lambda}(\beta_0) = \tilde{\Lambda}$ under the same tie-breaking convention used in Corollary 2. Since $P_{\beta_T}(E_T) \rightarrow 1$, the event-transfer lemma 19 transfers the fixed-order local limit to the unknown-order statistic. $\square$

Remark 9 (Proof audit for Theorem 7). The local theorem has two separate pieces. Lemma 21 proves the noncentral limit at the true partition from the calibrated local drift $g_{t,T}$, the local denominator stability, and the scalar EL expansion. Proposition 4 then moves the result from Λ_0 to $\tilde{\Lambda}$ and supplies local order consistency for unknown-order transfer.

9 Multiple-Break Extension Sketch

The main estimated-partition Wilks theorem is deliberately one-break. For a fixed number $q_0 > 1$, the natural extension is a telescoping argument rather than a new empirical-likelihood expansion. For a local candidate $\Lambda = (k_1, \dots, k_{q_0})$, define

$$\Lambda^{(0)} = \Lambda_0,$$

and

$$\Lambda^{(j)} = (k_1, \dots, k_j, k_{j+1,0}, \dots, k_{q_0,0}), \quad j = 1, \dots, q_0.$$

Then

$$\mathbf{M}_{K,\Lambda} - \mathbf{M}_{K,\Lambda_0} = \sum_{j=1}^{q_0} \{\mathbf{M}_{K,\Lambda^{(j)}} - \mathbf{M}_{K,\Lambda^{(j-1)}}\}.$$

Each summand moves one boundary between two long adjacent segments. If the true breaks are separated and q_0 is fixed, the one-break boundary, coefficient-update, local-bias, and projection-noise arguments can be applied to each summand, with only fixed multiplicative constants. The local candidate set has cardinality $O(r_T^{q_0})$, so the uniform projection bounds replace $\log(1 + r_T)$ by $q_0 \log(1 + r_T) = O(\log r_T)$.

This sketch is not used in the main theorem. A paper that claims general fixed- q_0 estimated-partition Wilks should extend the proof in the following order.

1. **Known multiple-break Wilks.** First keep Λ_0 known and fixed with fixed q_0 . The existing block-projection algebra already covers this case once the Gram, variance, convex-hull, and sieve conditions are stated uniformly over the known blocks.
2. **Fixed-order estimated bridge.** Next hold q_0 fixed and prove the componentwise bridge

$$\max_{1 \leq j \leq q_0} |\hat{k}_j - k_{j,0}| = O_p(r_T), \quad S_T(\hat{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\hat{\Lambda}) - V_T(\Lambda_0) = o_p(1).$$

This is where the telescoping display above must become a lemma, including the maximum-score condition for the scalar empirical-likelihood expansion.

3. **Multi-boundary localization.** Prove localization under fixed q_0 , fixed $\bar{q}$, minimum spacing, minimum jump size Δ_T , and non-overlapping Cr_T neighborhoods around the true breaks. The search-uniform stochastic terms should still be of order $R_T = K + \log T + Ta_K^2$ because q_0 is fixed.
4. **Unknown-order consistency.** Use the same penalty

$$\text{pen}_T(q, K) = q\kappa_T R_T.$$

Underfitted models must lose order $T\Delta_T^2$, while overfitted models may gain only $O_p(R_T)$. This is the multi-break analogue of Theorem 5.

5. **Final multi-break Wilks theorem.** Only after the fixed-order bridge and order-consistency proof are complete should the paper state

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Until those steps are written as full proofs, multi-break computations are exploratory diagnostics rather than main-theorem evidence.

10 End-to-End Assumption Audit

This final audit records the rebuilt paper architecture. The main theorem is not a general martingale, general-multiple-break result. It is a conditional- on-design, sub-Gaussian, one-break estimated-partition Wilks theorem, with unknown-order selection allowed over $q = 0, 1, \dots, \bar{q}$ when the true order is $q_0 = 1$.

Proof block	Required inputs	What it delivers
Conditional-design environment	Conditional independence, mean zero, bounded variance, and sub-Gaussian tails given $\mathcal{D}_T$	Full-sample residualized scores and selected partitions are design-measurable, avoiding predictable-score assumptions
Known-partition Wilks	Variance convergence, nondegeneracy, score envelope, convex hull, and sieve feasibility	$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2$ for fixed known q_0
One-break segmentation	Global Gram stability, deterministic profile identification, $R_T = K + \log T + Ta_K^2$, and $r_T = R_T/\Delta_T^2 = o(\sqrt{T})$	$ \hat{k} - k_0 = O_p(r_T)$ under $q_0 = 1$
Estimated-break bridge	Boundary size, local bias stability, local full rank, search-uniform projection noise, and bridge rates	$S_T(\hat{k}) - S_T(k_0) = o_p(1)$ and $V_T(\hat{k}) - V_T(k_0) = o_p(1)$
Unknown order	$\text{pen}_T(q, K) = q\kappa_T R_T$, $\kappa_T \rightarrow \infty$, and $\kappa_T R_T = o(T\Delta_T^2)$	$\hat{q}(\beta_0) = 1$ with probability tending to one
Local alternatives	$\beta_T = \beta_0 + h/\sqrt{T}$, $\Gamma_T \rightarrow_p \Gamma$, $\eta_T^2 \rightarrow_p \eta^2$, and the local bridge	Noncentral limit $\chi_1^2(h^2\Gamma^2/\eta^2)$
Multiple breaks	Telescoping over boundaries plus uniform multi-boundary localization	Extension sketch only; not part of the main theorem

Audit conclusion. The rebuilt manuscript now has one probabilistic spine. Conditional on $\mathcal{D}_T$, the EL score weights and the break search are fixed-design objects, so the stochastic tools are conditional CLT, conditional LLN, and finite-search sub-Gaussian concentration. The price is explicit: the main estimated-break Wilks and unknown-order Wilks results are stated for one nuisance break. A general fixed- q_0 claim requires the telescoping extension in Section 9 to be written as a full proof.

11 Closure Checklist After Editor–Author Loop

The harsh-editor objections addressed in this rebuild are recorded as closure checks rather than future work.

Editor objection	Author repair	Location
The stochastic environment mixed martingale scores with full-design break search	Replaced it by conditional independence and sub-Gaussian tails given $\mathcal{D}_T$	Assumption 1
The residualized instrument was not predictable	Made all projected scores design-measurable conditional-on- $\mathcal{D}_T$ objects	Assumption 2
The order theorem needed $R_T = o(T\Delta_T^2)$ and a concrete penalty	Set $R_T = K + \log T + Ta_K^2$, $r_T = R_T/\Delta_T^2$, and $\text{pen}_T(q, K) = q\kappa_T R_T$	Assumptions 16 and 17
The underfit RSS gap was implicit	Added an explicit zero-break comparison with $cT\Delta_T^2 - O_p(R_T)$	Assumption 16
The random-index projection bounds were pointwise	Added search-uniform T -norm and sup-norm projection-noise bounds	Lemma 14
The segmentation module still looked asserted rather than verified	Added a conditional-concentration proof of the uniform RSS expansion and order comparisons	Proposition 2
The score-bias condition was still imposed without a primitive route	Added a segmentwise orthogonality and smooth residualized-instrument verification	Assumption 7 and Lemma 2
Convex hull was a black-box EL condition	Added a primitive sign-mass condition and convex-hull verification lemma	Assumption 10 and Lemma 3
The endpoint theorem was implicit across several modules	Added a master one-break conditional-design Wilks theorem	Theorem 6
The rates might be mutually empty	Added a concrete smoothness/dimension regime with $s > d$	Corollary 1
Local power reused null-law bridges without saying why	Added local drift stability and local-law bridge/order transfer	Assumption 19 and Proposition 4
The estimated-partition theorem overclaimed multiple breaks	Restricted the main result to $q_0 = 1$ and moved fixed- q_0 to a telescoping sketch	Section 9

Final editorial status. Within the declared scope—conditional-on-design errors, one unknown nuisance break for estimated-partition Wilks, fixed $\bar{q}$, and the displayed rate regime—the workbook now states a coherent proof chain to known-break Wilks, estimated-break Wilks, and unknown-order transfer. The local noncentral result is also stated, but explicitly under the additional local drift-stability and local convex-hull inputs. The only material not claimed as a theorem is the general fixed-multiple-break extension, which is explicitly labeled as an extension sketch rather than part of the main result.